

The Impact of ESG, Bond, and Commodity Assets on Diversification in Traditional Mutual Fund Portfolios: A Cross-Market Study

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Abstract—In today's dynamic global financial landscape, investors are increasingly seeking diversification strategies that extend beyond traditional equity investments. This study investigates the impact of incorporating diverse asset classes—commodities, bonds, and Environmental, Social, and Governance (ESG) mutual funds—on the diversification and optimization of traditional mutual fund portfolios across international markets. ESG funds, selected for their dual promise of financial performance and alignment with sustainability objectives, address the growing investor demand for responsible investing options.

Employing daily global market data, portfolios are constructed using Modern Portfolio Theory (MPT), specifically leveraging Markowitz Mean-Variance optimization and Equal Risk Contribution (ERC) frameworks. Portfolio performance is rigorously assessed through traditional metrics such as Sharpe and Omega ratios, volatility, as well as advanced risk measures including Value at Risk (VaR), Conditional VaR (CVaR), Entropic VaR (EVar), and entropy.

Comprehensive backtesting spanning pre-COVID, COVID-19, and post-COVID market periods evaluates portfolio resilience during significant market disruptions. Additionally, Risk Parity and minimum variance strategies are deployed to further validate the robustness and adaptability of diversified portfolios under varying market conditions.

The results provide empirical evidence supporting the effectiveness of integrating commodities, bonds, and ESG assets into traditional mutual fund portfolios, highlighting significant diversification benefits and enhanced risk-adjusted performance. By emphasizing advanced risk measurement methodologies, this study contributes a nuanced understanding that can inform strategic asset allocation decisions, equipping investors and policymakers with tools to manage uncertainty and maintain resilience in volatile financial markets.

Keywords—*Modern Portfolio Theory (MPT), Sharpe Ratio, Omega Ratio, ESG Portfolio, Value at Risk (VaR), Conditional Value at Risk (CVaR), Entropy, Minimum Variance Spanning Test, Backtesting, Risk Management*

1. Introduction

In recent years, the global financial markets have experienced significant evolution, characterized by increased complexity and heightened volatility. These developments have posed challenges to traditional investment approaches, compelling investors to rethink portfolio diversification strategies. Diversification, a cornerstone of Modern Portfolio Theory (MPT), continues to play a critical role in mitigating unsystematic risk and optimizing risk-adjusted returns. Traditionally, investment portfolios have predominantly included equities and bonds. However, recurring financial crises, global pandemics such as COVID-19, and escalating geopolitical tensions have underscored the necessity for broader asset class inclusion. Consequently, investors have begun exploring alternative assets, notably commodities and Environmental, Social, and Governance (ESG) mutual funds, to enhance portfolio resilience and align investment decisions with sustainability objectives.

ESG investing, in particular, has gained considerable traction. ESG-focused mutual funds offer investors opportunities not only to achieve competitive financial returns but also to promote ethical, social, and environmental sustainability. Incorporating ESG factors into investment strategies is increasingly perceived as an effective tool for managing financial and non-financial risks, especially during volatile market conditions. Simultaneously, traditional alternative assets such as commodities and bonds remain vital components in portfolio diversification due to their historically low correlation with equities, enhancing risk management capabilities.

Risk assessment has gained prominence in contemporary investment decision-making, surpassing traditional metrics such as volatility, which provide limited insights into comprehensive risk-return dynamics. Advanced risk metrics, including Value at Risk (VaR), Conditional Value at Risk (CVaR), Entropic Value at Risk (EVar), and entropy, offer

deeper insights into portfolio performance under extreme market conditions and tail-risk scenarios. Additionally, refined performance measures like the Sharpe and Omega ratios facilitate nuanced evaluations of portfolio efficiency and effectiveness. Such multifaceted risk management approaches have become essential in navigating today's rapidly evolving and unpredictable financial landscapes, particularly when portfolios encompass diverse global asset classes and are tested by unprecedented events such as the COVID-19 pandemic.

Despite growing scholarly interest in sustainable investing and alternative asset integration, the existing literature demonstrates significant gaps concerning comprehensive analyses of multi-asset portfolios combining ESG mutual funds with traditional and alternative investments. Prior research often addresses financial performance or sustainability separately, neglecting integrated risk-return optimization under realistic and stress-induced market conditions. Furthermore, limited studies employ advanced optimization techniques or perform minimum variance spanning tests to investigate whether the inclusion of ESG, commodities, and bonds effectively expands the efficient frontier and enhances diversification in traditional portfolios.

This study seeks to bridge these literature gaps by systematically constructing diversified portfolios incorporating commodities, bonds, and ESG mutual funds, evaluating their performance using advanced optimization frameworks such as Markowitz Mean-Variance optimization and Equal Risk Contribution (ERC). It applies sophisticated risk measurement methodologies and conducts rigorous back-testing across distinct market periods—pre-COVID, COVID-19, and post-COVID—to rigorously examine portfolio resilience and robustness under various systemic shocks. Thus, this research contributes practical insights and methodologies for effectively integrating diverse asset classes, guiding both investors and policymakers toward enhanced portfolio resilience and informed asset allocation strategies.

2. Literature Review

Effective portfolio diversification, a cornerstone principle of Modern Portfolio Theory (MPT), has been extensively recognized for mitigating unsystematic risk and enhancing overall investment returns **markowitz1952portfolio**, **elton2009modern**. Traditionally, equity and

fixed-income securities have dominated asset allocation strategies. However, recent shifts in financial markets, characterized by increased volatility and systemic uncertainty, have led to expanded diversification strategies that integrate alternative assets such as commodities and Environmental, Social, and Governance (ESG) mutual funds **gorton2006facts**, **ammann2018esg**.

Commodities have consistently demonstrated hedging capabilities against inflationary pressures and market downturns, primarily due to their low correlation with traditional asset classes such as equities and bonds **bessembinder1992systematic**, **erdos2012role**. Bonds, especially sovereign and high-quality corporate bonds, have similarly been validated as essential stabilizing assets, offering counter-cyclical protection during periods of financial stress **ilmanen2003stock**, **campbell1997econometrics**. Concurrently, ESG investing has emerged prominently, driven by investor demand for portfolios that deliver financial returns while simultaneously aligning with sustainability goals. Empirical research indicates ESG funds not only provide competitive financial performance but also effectively manage risk, particularly through the mitigation of long-term systemic and idiosyncratic risks associated with sustainability-related issues **friede2015esg**, **clerckx2015esg**, **fattouh2020esg**.

Quantitative approaches in portfolio diversification typically employ Markowitz's Mean-Variance optimization framework, which remains foundational for balancing risk and return **markowitz1952portfolio**. Complementing this approach, Equal Risk Contribution (ERC) models have gained prominence for their ability to allocate risk uniformly across portfolio assets, thereby reducing concentration risks **maillard2010properties**. Recent methodological advancements incorporate sophisticated risk metrics such as Value at Risk (VaR), Conditional Value at Risk (CVaR), Entropic Value at Risk (EVaR), and entropy, providing nuanced assessments of tail risks and extreme market conditions **rockafellar2002conditional**, **ahmadi2012entropic**, **philippe2013entropy**. Furthermore, performance metrics such as Sharpe and Omega ratios offer refined evaluations beyond traditional volatility measures, delivering deeper insights into portfolio efficiency and risk-adjusted returns **sharp1966mutual**, **keating2002omega**,

caporin2014omega.

Despite substantial progress, notable research gaps persist. Existing literature typically isolates financial returns from ESG considerations, limiting the comprehensive understanding of integrated performance and risk profiles. Additionally, there is insufficient exploration of advanced optimization techniques, such as higher-order portfolio optimization and minimum variance spanning tests, necessary for robust assessments of portfolio diversification efficiency **kan2012tests**, **huberman1987mean**. Critically, the resilience and performance of multi-asset portfolios incorporating commodities, bonds, and ESG funds during extraordinary market disruptions—most notably the COVID-19 pandemic—are inadequately addressed.

This research aims to bridge these gaps by systematically evaluating diversified portfolios that integrate commodities, bonds, and ESG mutual funds. Employing an extensive methodological framework, this study leverages Mean-Variance, ERC, and higher-order optimization techniques. Rigorous backtesting across diverse market periods—pre-COVID, COVID-19, and post-COVID—further strengthens the analysis, ensuring the robustness and practical relevance of findings. Consequently, this research significantly contributes to existing literature and offers valuable insights for investors, asset managers, and policymakers focused on achieving sustainable and resilient portfolio strategies.

3. Data Source and Description

This study uses a comprehensive dataset obtained from Bloomberg Terminal, known globally for its reliability and extensive financial data coverage. The dataset includes daily closing prices of mutual funds from January 2005 to March 2025, covering significant market events such as the 2008 financial crisis and the COVID-19 pandemic. This long-term data allows for detailed analysis and testing of portfolio strategies under various market conditions.

Data from three major financial markets—the United States, India, and Europe—are analyzed. This approach provides valuable insights into how diversification strategies perform differently across various regions. Mutual funds were selected from several asset classes, ensuring the relevance and practicality of the analysis for modern investment strategies.

The mutual fund categories included in this study are:

- **Equity Mutual Funds:** Primarily focused on capital growth through investments in stocks.
- **Debt Mutual Funds:** Investments in fixed-income securities such as corporate bonds and government bonds, providing regular income and stability.
- **Commodity Mutual Funds:** Investments linked to commodities like gold, silver, and oil, offering protection against inflation and market downturns.
- **Bond and Treasury Bond Mutual Funds:** Investments in government and treasury securities of different maturities, adding stability to portfolios.
- **ESG Mutual Funds:** Investments in companies meeting Environmental, Social, and Governance criteria, combining financial goals with sustainable investing practices.

3.1. Portfolio Construction

Portfolio construction involves selecting and combining various asset classes to achieve effective diversification and optimal performance. In this study, five distinct types of portfolios were created for each market (U.S., India, and Europe) to evaluate the effectiveness of different asset combinations:

1. **Traditional Portfolio (Baseline):** Includes only equity and debt mutual funds, reflecting common investment practices.
2. **Traditional + Commodity Portfolio:** Adds commodity mutual funds to the baseline portfolio, assessing diversification through tangible assets.
3. **Traditional + ESG Portfolio:** Incorporates ESG mutual funds into the traditional portfolio to explore financial performance alongside sustainability.
4. **Traditional + Bond Portfolio:** Adds bond and treasury bond mutual funds to evaluate their effectiveness in stabilizing portfolio performance during market stress.
5. **Comprehensive Diversified Portfolio:** Combines all asset types—equities, debt, commodities, ESG, and bonds—to fully assess the benefits of comprehensive diversification.

Each type of portfolio was constructed separately for the U.S., Indian, and European markets, allow-

ing comparisons of portfolio performance across different economic environments. The portfolios were periodically rebalanced to maintain intended asset allocations. Additionally, data were carefully processed to ensure accuracy and consistency, including aligning trading days, handling missing data appropriately, and normalizing returns. This meticulous data preparation ensures reliable analysis results, useful for investors, asset managers, and policymakers.

4. Methodology

This study follows a structured, multi-step methodology based on **Modern Portfolio Theory (MPT)** and supported by advanced optimization and risk assessment techniques. By combining traditional financial models with modern computational tools, we analyze how adding equity, commodity, bond, and ESG mutual funds affects portfolio diversification across the U.S., India, and Europe.

Portfolio diversification analysis Methodology Flowchart

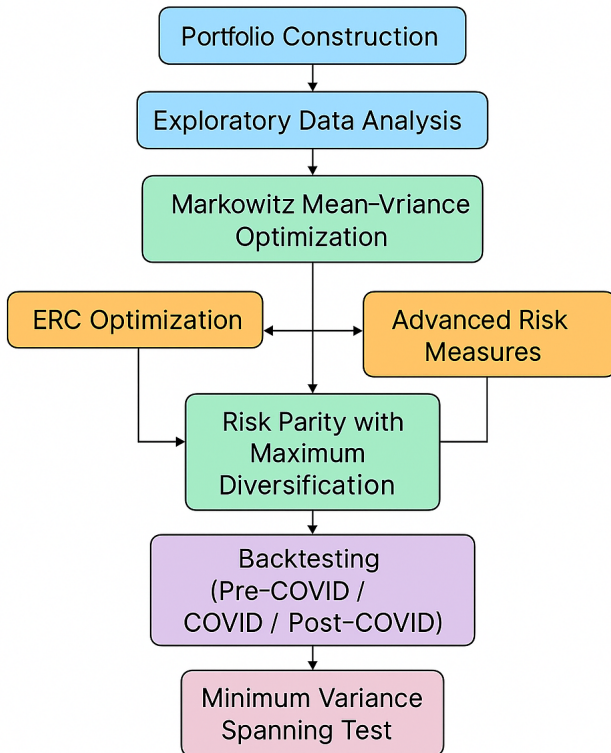


Figure 1. Portfolio Analysis Flowchart: Overview of Methodological Pipeline

The data (mostly were from 2005–2025) from each category were meticulously merged by matching trading dates to ensure temporal consistency. Following this, comprehensive exploratory data analysis (EDA) was performed to understand data patterns, detect anomalies, and prepare the dataset for portfolio optimization.

5. Methodology

5.1. Markowitz Mean-Variance Optimization

Markowitz Mean-Variance Optimization serves as the foundational framework of Modern Portfolio Theory (MPT), aimed at identifying the most efficient portfolio for a given level of risk or expected return. This method constructs the efficient frontier, representing a set of portfolios that offer the highest expected return for a defined level of risk.

The optimization problem is formulated as:

$$\min_{\mathbf{w}} \mathbf{w}^T \Sigma \mathbf{w} \quad \text{s.t.} \quad \begin{aligned} \mathbf{w}^T \boldsymbol{\mu} &= \mu_p, \\ \sum w_i &= 1, \quad w_i \geq 0 \end{aligned}$$

Here, $\boldsymbol{\mu}$ is the vector of expected returns, Σ is the covariance matrix of asset returns, and μ_p is the target return of the portfolio. The constraints ensure that the portfolio is fully invested (weights sum to 1) and long-only (no short selling).

This optimization seeks to minimize portfolio variance while achieving a specific expected return, providing a quantitative approach to balancing risk and return in asset allocation.

Performance Evaluation Metrics:

To assess the effectiveness of the optimized portfolios, the following key metrics are used:

- **Expected Return ($\mathbb{E}[R_p]$):**

$$\mathbb{E}[R_p] = \mathbf{w}^T \boldsymbol{\mu}$$

Represents the portfolio's anticipated average return, calculated as the weighted sum of the individual asset returns.

- **Volatility (σ_p):**

$$\sigma_p = \sqrt{\mathbf{w}^T \Sigma \mathbf{w}}$$

Measures the overall risk of the portfolio by capturing the standard deviation of returns.

- **Sharpe Ratio:**

$$\text{Sharpe} = \frac{\mathbb{E}[R_p] - R_f}{\sigma_p}$$

Evaluates risk-adjusted performance by indicating the excess return per unit of risk relative to a risk-free asset.

- **Omega Ratio:** The Omega Ratio evaluates a portfolio's performance by comparing the probability-weighted gains to the probability-weighted losses relative to a threshold return θ . Unlike traditional ratios that assume normally distributed returns, the Omega Ratio considers the entire distribution, making it particularly useful for portfolios with non-normal or asymmetric return profiles.

$$\Omega(\theta) = \frac{\int_{\theta}^{\infty} (1 - F(x)) dx}{\int_{-\infty}^{\theta} F(x) dx}$$

Here, $F(x)$ represents the cumulative distribution function (CDF) of portfolio returns. A higher Omega Ratio indicates a greater likelihood of achieving favorable outcomes relative to downside risks.

- **Beta (β):** Beta measures the sensitivity of a portfolio's returns to changes in the overall market. It captures the extent to which systematic risk influences portfolio performance. A beta greater than 1 implies higher volatility than the market, while a beta less than 1 suggests the portfolio is less volatile than the market.

$$\beta = \frac{\text{Cov}(R_p, R_m)}{\text{Var}(R_m)}$$

Where:

- R_p : Return of the portfolio
- R_m : Return of the market benchmark
- $\text{Cov}(R_p, R_m)$: Covariance between portfolio and market returns
- $\text{Var}(R_m)$: Variance of market returns .

We analyzed the performance metrics like Sharpe Ratio and Omega Ratio under two different portfolio strategies: one with equal weight allocation and the other with optimized weight allocation. This analysis was conducted on both the original return data and the normalized return data to examine how weighting schemes and data scaling affect portfolio performance.

5.2. Equal Risk Contribution (ERC) Optimization

Ensures that each asset contributes equally to total portfolio risk, reducing concentration risk.

Equal Risk Contribution (ERC) Optimization is a portfolio strategy that allocates asset weights such that each asset contributes equally to the total portfolio risk. Unlike return-focused models, ERC emphasizes balanced risk distribution, leading to more diversified and stable portfolios, especially under market uncertainty.

Risk Contribution of asset i :

$$RC_i = w_i \cdot \frac{(\Sigma \mathbf{w})_i}{\sqrt{\mathbf{w}^T \Sigma \mathbf{w}}}$$

Objective:

$$\min_{\mathbf{w}} \sum_{i=1}^N \sum_{j=1}^N (RC_i - RC_j)^2 \quad (1)$$

subject to $\sum w_i = 1, \quad w_i \geq 0$

5.3. Advanced Risk Measures

These techniques provide a more accurate understanding of tail risk and portfolio uncertainty.

- **Value at Risk (VaR):** VaR is a widely used risk measure that estimates the maximum potential loss in portfolio value over a given time horizon, at a specified confidence level. It provides a threshold below which the portfolio's losses are not expected to fall with a certain probability.

$$\text{VaR}_{\alpha} = -\inf\{x \mid P(R_p \leq x) \geq \alpha\}$$

Estimated using:

- **Parametric:** $\text{VaR}_{\alpha} = z_{\alpha} \cdot \sigma_p$
- **Historical Simulation**
- **EWMA:** $\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) R_{t-1}^2$

- **Conditional Value at Risk (CVaR):** CVaR, also known as Expected Shortfall, measures the average loss of a portfolio in the worst-case scenarios beyond the Value at Risk (VaR) threshold. It provides a more comprehensive view of tail risk, especially under extreme market conditions.

$$\text{CVaR}_{\alpha} = \mathbb{E}[R_p \mid R_p \leq \text{VaR}_{\alpha}]$$

- **Entropic Value at Risk (EVaR):** EVaR is a coherent risk measure that provides an upper bound for both Value at Risk (VaR) and Conditional Value at Risk (CVaR). It is based on the concept of entropy and captures tail risk by considering the worst-case expected loss under

exponential decay, making it especially useful for portfolios with heavy-tailed distributions.

$$\text{EVaR}_\alpha(R_p) = \inf_{z>0} \left\{ \frac{1}{z} \log \left(\frac{\mathbb{E}[e^{-zR_p}]}{\alpha} \right) \right\}$$

- **Semi-Entropy:** Semi-Entropy is a downside risk measure that quantifies uncertainty only in the negative return region. Unlike full entropy, it focuses on losses, making it relevant for risk-averse investors.

$$\text{SemiEntropy} = - \sum_{r_i < \bar{r}} p_i \log(p_i)$$

Where r_i are returns below the mean $\bar{r}$, and p_i are their probabilities. Higher semi-entropy implies greater unpredictability in losses.

- **Cross Entropy:** Cross Entropy quantifies the difference between the actual portfolio allocation (P) and a target or reference distribution (Q), useful in comparing divergence from an ideal weighting scheme.

$$CE(P||Q) = - \sum_{i=1}^N P_i \log Q_i$$

- **Semi-Variance:** Semi-Variance focuses only on the downside deviation of returns, measuring the average squared loss when returns fall below the mean.

$$\text{SemiVar} = \frac{1}{T} \sum_{i=1}^T \min(R_i - \mu, 0)^2$$

- **Semi-Kurtosis:** Semi-Kurtosis captures the extremity of downside tail events by measuring the fourth power of negative deviations from the mean, emphasizing severe losses.

$$\text{SemiKurt} = \frac{1}{T} \sum_{i=1}^T \min(R_i - \mu, 0)^4$$

5.4. Risk Parity with Maximum Diversification

This approach aims to construct portfolios where each asset contributes equally to overall portfolio risk, while also maximizing diversification.

Definition: Risk Parity focuses on balancing the risk contributions of all assets rather than their capital weights. This ensures that no single asset dominates the portfolio's risk profile. When combined with the Maximum Diversification principle—which aims to maximize the ratio of weighted

average volatility to total portfolio volatility—the resulting portfolio enhances stability and diversification.

$$\max_{\mathbf{w}} \frac{\mathbf{w}^\top \boldsymbol{\sigma}}{\sqrt{\mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w}}} \quad \text{subject to} \quad \sum w_i = 1, \quad w_i \geq 0$$

Where:

- $\boldsymbol{\sigma}$: Vector of individual asset volatilities
- $\boldsymbol{\Sigma}$: Covariance matrix of asset returns
- $\mathbf{w}$: Vector of portfolio weights

This formulation leads to portfolios that are both well-diversified and risk-balanced, offering improved resilience in volatile market environments.

5.5. Backtesting Framework

Backtesting evaluates the robustness and performance consistency of portfolio strategies under various historical market conditions.

To ensure comprehensive evaluation, portfolios are tested across three major economic regimes:

- **Pre-COVID (2005–2019):** A period of relative market stability and growth.
- **COVID-19 Crisis (2020–2021):** High volatility and uncertainty due to the global pandemic.
- **Post-COVID Recovery (2022–2025):** Economic rebound and structural shifts in financial markets.

Performance Metrics Computed:

- **Compound Annual Growth Rate (CAGR):** CAGR measures the mean annual growth rate of portfolio value over a period, assuming compounding.

$$\text{CAGR} = \left(\frac{V_T}{V_0} \right)^{1/T} - 1$$

- **Average Excess Return:** Represents the average return of the portfolio above the risk-free rate, capturing relative performance.

$$\bar{R}_{\text{excess}} = \frac{1}{T} \sum_{t=1}^T (R_t - R_f)$$

5.6. Minimum Variance Spanning Test (MVST)

The Minimum Variance Spanning Test (MVST) statistically evaluates whether adding a new asset class

(e.g., ESG, bond, or commodity mutual funds) improves portfolio efficiency. This is done by checking whether the expanded asset universe shifts the minimum-variance frontier outward.

Efficient Frontier Comparison (Graphical Method)

The efficient frontier represents the set of optimal portfolios that either maximize expected return for a given level of risk or minimize risk for a given return. Graphically, we compare two frontiers:

- P_T : The base portfolio without the new asset class.
- P_T^* : The extended portfolio with the new asset class.

The null and alternative hypotheses are:

H_0 : The min-variance frontier of P_T spans that of P_T^*

H_1 : The frontier of P_T^* lies outside that of P_T

If H_0 is rejected, we conclude that the new asset class improves portfolio efficiency. This can be visually confirmed by plotting the two frontiers and observing whether P_T^* lies entirely outside P_T .

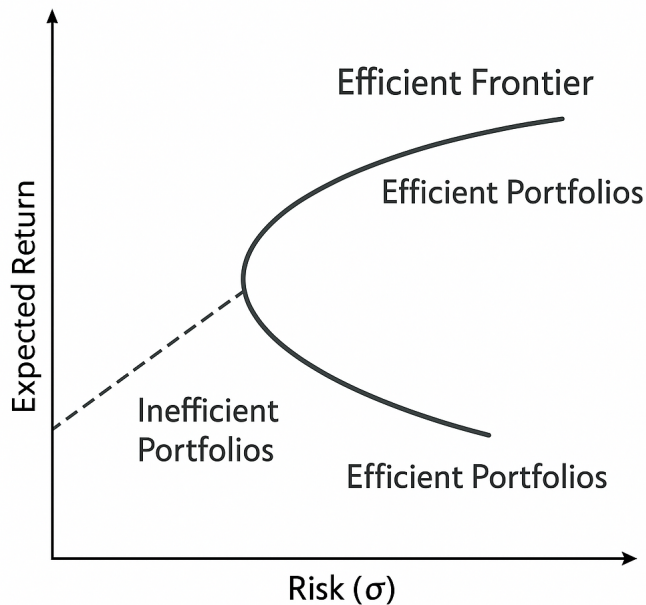


Figure 2. Efficient Frontier Comparison

Likelihood Ratio (LR) Test

To formally test whether the inclusion of new assets statistically improves the minimum-variance portfolio, we apply the Likelihood Ratio (LR) test. This

method compares the minimum variances of the restricted (original) and unrestricted (augmented) portfolios.

Let:

- Σ_R : Covariance matrix of the base portfolio
- Σ_X : Covariance matrix of the extended portfolio
- T : Number of time observations

The minimum variance for a fully invested portfolio is:

$$\sigma_{\min}^2 = \frac{1}{\mathbf{1}^\top \Sigma^{-1} \mathbf{1}}$$

The LR test statistic is then:

$$\Lambda = T \cdot \ln \left(\frac{\sigma_{\min,R}^2}{\sigma_{\min,X}^2} \right) = T \cdot \ln \left(\frac{\mathbf{1}^\top \Sigma_X^{-1} \mathbf{1}}{\mathbf{1}^\top \Sigma_R^{-1} \mathbf{1}} \right)$$

Under the null hypothesis, $\Lambda \sim \chi^2(k)$, where k is the number of additional assets added. If Λ exceeds the critical value from the chi-squared distribution, we reject H_0 , indicating a statistically significant improvement in diversification.

Lagrange Duality Matrix Optimization

The minimum-variance portfolio problem can also be formulated using Lagrangian optimization. This approach provides the exact weight vector of the optimal portfolio and confirms whether the new assets shift the efficient frontier.

Primal Optimization:

$$\min_{\mathbf{w}} \mathbf{w}^\top \Sigma \mathbf{w} \quad \text{subject to} \quad \mathbf{w}^\top \mathbf{1} = 1$$

Lagrangian Form:

$$\mathcal{L}(\mathbf{w}, \lambda) = \mathbf{w}^\top \Sigma \mathbf{w} - \lambda(\mathbf{w}^\top \mathbf{1} - 1)$$

Solving First-Order Conditions:

$$\nabla_{\mathbf{w}} \mathcal{L} = 2\Sigma \mathbf{w} - \lambda \mathbf{1} = 0 \quad \Rightarrow \quad \mathbf{w}^* = \frac{\lambda}{2} \Sigma^{-1} \mathbf{1}$$

Using the constraint $\mathbf{w}^{*\top} \mathbf{1} = 1$, we solve for λ :

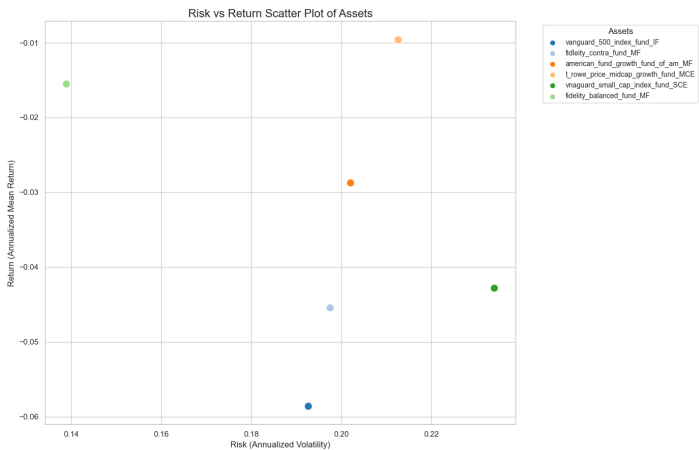
$$\lambda = \frac{2}{\mathbf{1}^\top \Sigma^{-1} \mathbf{1}}, \quad \Rightarrow \quad \mathbf{w}^* = \frac{\Sigma^{-1} \mathbf{1}}{\mathbf{1}^\top \Sigma^{-1} \mathbf{1}}$$

6. Results and Discussion

6.1. US Market

6.1.1. Traditional Portfolio

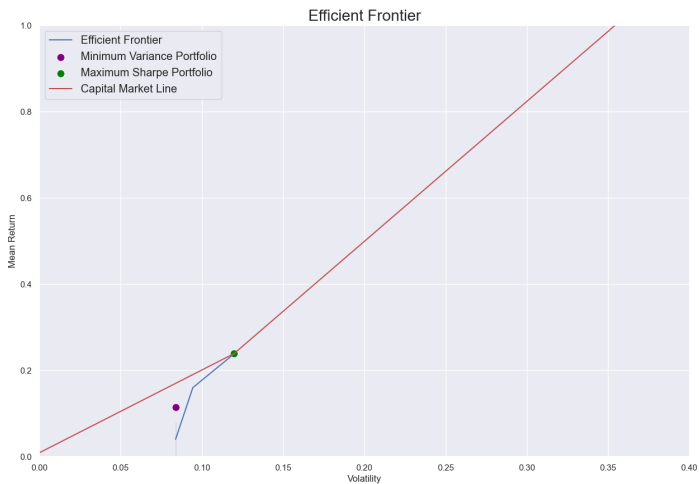
The scatter plot presents the risk-return profiles of various mutual funds. Conservative investment options such as the Vanguard 500 Index Fund (IF) and the Fidelity Balanced Fund (MF) demonstrate lower



Risk vs Return Scatter Plot of Assets

levels of both risk and return. In contrast, more aggressive strategies, exemplified by the American Fund Growth Fund of AM (MF) and the Vanguard Small Cap Index Fund (SCE), offer elevated risk and potential returns, appealing to investors seeking growth.

Efficient Frontier



Efficient Frontier of the Optimized Portfolio

The Efficient Frontier represents the set of portfolios that achieve the maximum expected return for a given level of risk. Key components of the plot include:

- **Efficient Frontier (Blue Line):** Denotes optimal portfolios offering the highest return for each unit of risk.
- **Capital Market Line (Red Line):** Illustrates the combinations of risky assets and a risk-free asset providing the best risk-return trade-off.
- **Minimum Variance Portfolio (Purple Dot):** The portfolio with the lowest total volatility.

- **Maximum Sharpe Portfolio (Green Dot):** The portfolio offering the highest risk-adjusted return.

Insights from the frontier include:

- The Minimum Variance Portfolio is ideal for risk-averse investors prioritizing stability over high returns.
- Portfolios along the frontier allow investors to balance return expectations with acceptable risk levels.
- The Maximum Sharpe Portfolio delivers the most efficient trade-off, maximizing performance relative to risk.

This framework aids in identifying optimal asset allocations aligned with specific investment goals.

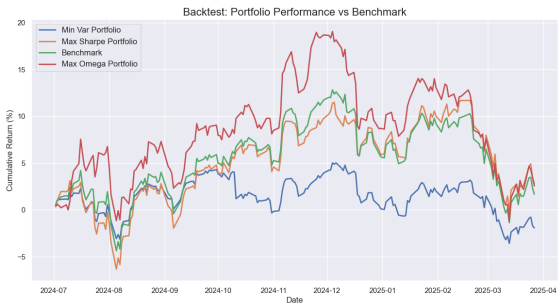
Omega Ratio Comparison

Table 1. Omega Ratio Comparison

Portfolio Type	Ratio
Baseline Equal-weighted	0.0728
Optimized Max Omega	0.1067
Improvement	0.0338

The Omega ratio improved from 0.0728 to 0.1067 through optimization, reflecting a gain of 0.0338. This enhancement indicates superior risk-adjusted performance, with the optimized portfolio exhibiting a higher probability of positive returns relative to losses.

Backtesting Portfolio Performance vs Benchmark



Backtest: Portfolio Performance vs Benchmark

- **Minimum Variance (Blue):** Demonstrates stable performance with low volatility but limited upside potential.

- **Max Sharpe (Orange):** Outperforms Minimum Variance through higher returns while maintaining a reasonable risk level.
- **Max Omega (Red):** Generates the highest cumulative return, maximizing reward relative to risk.
- **Benchmark (Green):** Offers average performance, lagging behind all optimized strategies.

The backtest reveals that optimization based on Sharpe and Omega ratios consistently outperforms the benchmark, with the Max Omega Portfolio exhibiting the most favorable performance.

Sharpe vs Omega Ratio Analysis

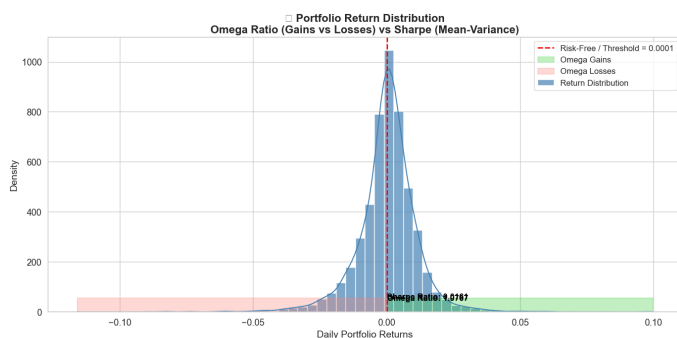


Figure 6. Portfolio Return Analysis: Sharpe vs Omega Ratios

- **Top-Left (Histogram & KDE):** Displays returns centered around the mean. The green area denotes gains above the risk-free rate, emphasizing upside strength.
- **Top-Right (Boxplot):** Shows symmetric spread with moderate outliers, which can influence Sharpe ratios.
- **Bottom-Left (CDF):** Steep slope suggests high likelihood of achieving returns above the risk-free benchmark.
- **Bottom-Right (Violin Plot):** Highlights dense clustering and slight left skew; the Omega ratio effectively captures this tail behavior.

While the Sharpe ratio measures efficiency in mean-variance terms, the Omega ratio accounts for asymmetry and extreme outcomes, offering a more comprehensive view of portfolio risk-return characteristics.

Portfolio Optimization: Risk Parity, Maximum Diversification, and Minimum Variance

Table 2. Portfolio Optimization Results: Risk Parity, Maximum Diversification, Minimum Variance

Metric	Equal Weight	Risk Parity	Min. Variance	Max Divers
AER	7.24%	7.24%	7.24%	6.54%
SD	18.48%	18.48%	18.48%	18.04%
SR	0.28	0.28	0.28	0.25
CR	5.68%	5.68%	5.68%	5.03%
Beta	1.00	1.00	1.00	0.97

- **AER** – Average Excess Return
- **SD** – Standard Deviation
- **SR** – Sharpe Ratio
- **CR** – Compound Return
- **Beta** – Market Sensitivity Measure

- **Equal Weighted:** Delivers moderate return and volatility, offering broad exposure.
- **Risk Parity:** Matches Equal Weighted in key metrics, emphasizing risk-balanced allocation.
- **Minimum Variance:** Achieves comparable performance with reduced volatility focus.
- **Max Divers.:** Offers the lowest risk level but at the cost of lower returns and Sharpe ratio.
- **Market (VW):** Exhibits the highest return and risk; best suited for return-maximizing investors.

Portfolio Risk Metrics Comparison

Table 3. Risk Metrics Before and After Optimization

Metric	
Entropy	2.171757
Semi-Variance	0.000109
CVaR	-0.034655
EVaR	-0.000278
Cross-Entropy	6.108082
Semi-Entropy	2.279704
Semi-Kurtosis	23.592167

The optimized portfolio demonstrates:

- A decrease in **entropy**, suggesting enhanced predictability.
- Improvement in **CVaR and EVaR**, highlighting better tail-risk control.

- Slight increases in **semi-variance**, **semi-entropy**, and **semi-kurtosis**, implying greater attention to downside exposure.
- Elevated **cross-entropy**, indicating notable deviation from initial allocations due to optimization.



Figure 8. Portfolio Risk-Return Metrics Across Periods

The bar chart comparison indicates:

- **Sharpe Ratio and Return:** Peaked during the COVID period due to market rebounds and policy support.
- **Volatility:** Persisted across all periods, reflecting ongoing macroeconomic uncertainty.
- **Omega Ratio:** Increased consistently, peaking Post-COVID, suggesting improved gain-loss asymmetry.

These results highlight the importance of adaptive strategies and risk-sensitive metrics like the Omega ratio under varying economic conditions.

COVID-19 Period Analysis

To assess the impact of macroeconomic conditions, portfolio performance was analyzed across three distinct periods:

- **Pre-COVID (2018–2019):** Characterized by market stability and consistent returns.
- **COVID (2020–2021):** Period of elevated volatility and disruption due to the global pandemic.
- **Post-COVID (2022–Present):** Marked by economic recovery, inflation, and shifting policy dynamics.

Efficient Frontier and Risk-Return Shifts: Pre-COVID vs. COVID vs. Post-COVID

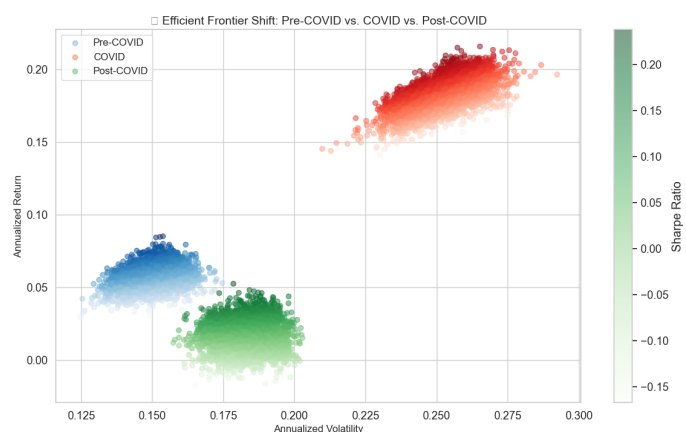


Figure 7. Efficient Frontier Shift: Pre-COVID, COVID, and Post-COVID Periods

The efficient frontier plots reveal that during COVID, portfolios experienced higher returns alongside greater volatility, leading to improved Sharpe ratios. In the Post-COVID period, both returns and risk-adjusted performance declined compared to the COVID phase.

6.1.2. Traditional + Commodity Portfolio

To assess the impact of commodity inclusion, we evaluate the performance of a traditional portfolio versus a commodity-augmented portfolio through three core analyses: the shift in the efficient frontier, return distribution comparison using a likelihood ratio test, and optimization via Lagrange duality matrix.

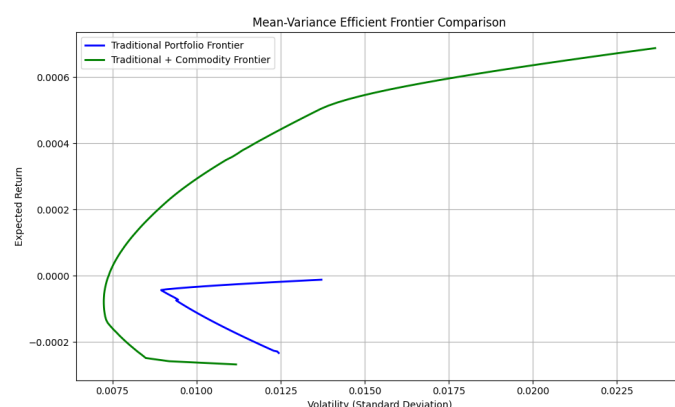


Figure 9. Efficient Frontier: Traditional vs. Traditional + Commodity

The efficient frontier corresponding to the portfolio with commodities (green) lies above and to the

left of the traditional frontier (blue), indicating improved portfolio efficiency. This shift confirms that the inclusion of commodities enhances returns for a given level of risk, or alternatively, reduces risk for a targeted level of return—highlighting the diversification benefit.

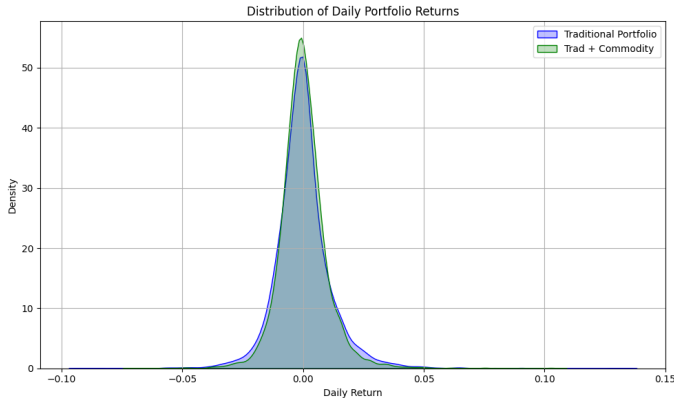


Figure 10. Return Distribution: Traditional vs. Traditional + Commodity

The return distribution of the combined portfolio is more concentrated and centered, reflecting enhanced return stability and reduced downside deviations. This pattern is indicative of improved performance consistency.

Lagrange Duality Test Results

The minimum variance derived from the Lagrange duality optimization approach reveals the following:

- **Traditional Portfolio:** 0.000058
- **Portfolio with Commodities:** 0.000037
- **Variance Reduction:** 0.000020

The following improvements are observed upon incorporating commodities:

- The efficient frontier exhibits a significant outward shift.
- Return distributions show greater centrality and lower downside risk.
- Minimum variance is notably reduced.

the inclusion of commodities substantially improves portfolio optimization outcomes by enhancing diversification and reducing risk exposure, leading to superior risk-adjusted returns.

Interpretation: The commodity-augmented portfolio demonstrates a notable improvement in risk-adjusted performance, as reflected by the increased

Table 4. Performance Metrics: Traditional vs. Traditional + Commodity Portfolio (After Optimization)

Metric	Trad. Portfolio	Trad + Comm Portfolio
Sharpe Ratio	0.4463	0.6546
Sharpe Expected Return	0.0955	0.0983
Sharpe Volatility	0.1916	0.1349
Omega Ratio	1.1024	1.1417
Omega Expected Return	0.0955	0.0983
Omega Volatility	0.1916	0.1349

Sharpe and Omega ratios. Both expected return and volatility metrics improve slightly, underscoring the value of commodities in enhancing diversification while optimizing return potential.

Table 5. Risk Metrics: Traditional vs. Traditional + Commodity Portfolio (After Optimization)

Metric	Trad Portfolio	Trad+Comm Portfolio
Entropy	2.171757	2.210298
Semi-Variance	0.000109	0.000220
CVaR	-0.034655	-0.047156
EVaR	-0.000278	-0.000412
Cross-Entropy	6.108082	7.626251
Semi-Entropy	2.279704	2.646351
Semi-Kurtosis	23.592167	13.221528

Interpretation: The post-optimization risk metrics reveal enhanced portfolio structure due to commodity inclusion. Higher entropy and cross-entropy values indicate broader diversification and allocation shifts. Lower CVaR and EVaR suggest improved control over extreme losses, while the marked reduction in semi-kurtosis points to a decreased occurrence of outlier returns. Collectively, these outcomes validate the role of commodities in stabilizing portfolio risk and enhancing return consistency.

6.1.3. Traditional + ESG Portfolio

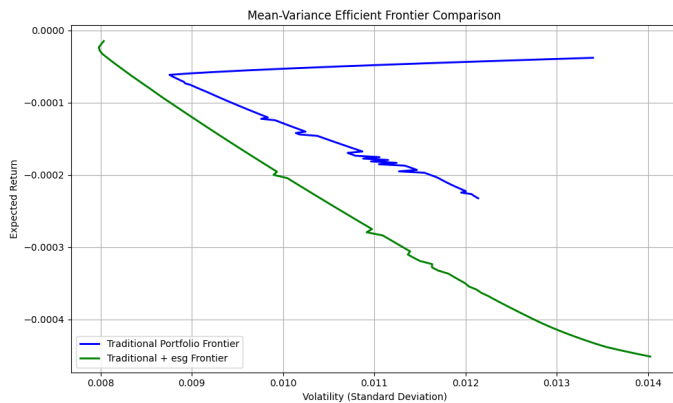


Figure 11. Return Distribution: Traditional vs. Traditional + ESG Portfolio

The return distribution of the ESG-augmented portfolio appears narrower and more peaked compared to the traditional portfolio. This suggests a greater concentration of returns around the mean and modest improvement in downside risk control, indicating enhanced return stability.

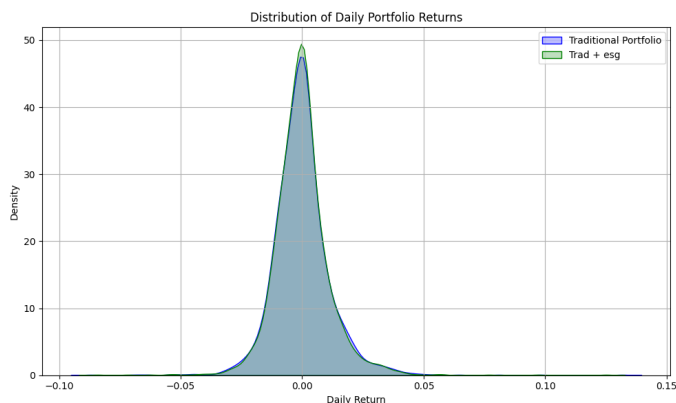


Figure 12. Efficient Frontier: Traditional vs. Traditional + ESG Portfolio

Contrary to expectations, the efficient frontier of the Traditional + ESG portfolio (green line) lies below that of the traditional portfolio (blue line), reflecting lower expected returns at similar risk levels. This result implies that, within the mean-variance framework applied to this dataset, ESG integration does not contribute to improved return optimization.

Lagrange Duality Test Results

- **Traditional Portfolio:** 0.000073
- **Traditional + ESG:** 0.000040
- **Difference:** 0.000033

The reduction in minimum variance highlights the risk-minimizing effect of ESG integration. Although it narrows the return distribution and enhances downside risk control, this improvement is not accompanied by a corresponding gain in return potential or efficient frontier elevation.

Overall, ESG integration enhances portfolio stability and risk containment but does not significantly improve return optimization outcomes within the applied optimization model.

Table 6. Performance Metrics After Optimization: Traditional vs. Traditional + ESG Portfolio

Metric	Traditional Portfolio	Traditional + ESG Portfolio
Sharpe Ratio	0.4463	0.6915
Sharpe Expected Return	0.0955	0.1622
Sharpe Volatility	0.1916	0.2201
Omega Ratio	1.1024	1.1423
Omega Expected Return	0.0955	0.1622
Omega Volatility	0.1916	0.2201

Interpretation: The inclusion of ESG assets improves risk-adjusted performance, as evidenced by higher Sharpe and Omega ratios. Although volatility slightly increases, the portfolio achieves a notably higher expected return. This reflects a more sustainable investment structure without compromising performance, supporting ESG's role in strategic allocation.

Table 7. Risk Metrics After Optimization: Traditional vs. Traditional + ESG Portfolio

Metric	Traditional Portfolio	Traditional + ESG Portfolio
Entropy	2.171757	2.485813
Semi-Variance	0.000109	0.000116
CVaR	-0.034655	-0.036036
EVaR	-0.000278	-0.000220
Cross-Entropy	6.108082	5.350136
Semi-Entropy	2.279704	2.311896
Semi-Kurtosis	23.592167	27.746135

Interpretation: Post-optimization risk metrics show improved entropy and slightly better CVaR and EVaR, indicating enhanced diversification and

stronger tail-risk mitigation. However, the increase in semi-kurtosis suggests a higher frequency of extreme

6.1.4. Traditional + Bond Portfolio

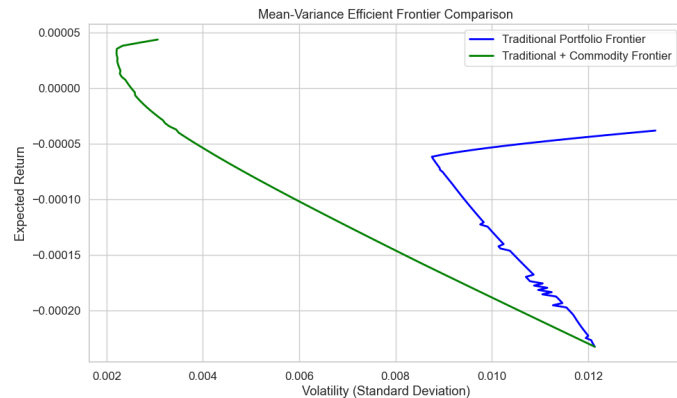


Figure 13. Return Distribution: Traditional vs. Traditional + Bond Portfolio

The return distribution of the Traditional + Bond Portfolio appears taller and narrower compared to the traditional-only portfolio. This shape indicates a tighter concentration of returns around the mean, signifying reduced volatility and improved downside risk containment.

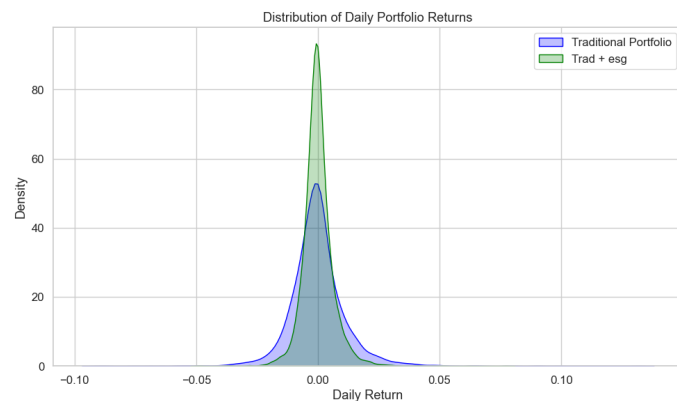


Figure 14. Efficient Frontier: Traditional vs. Traditional + Bond Portfolio

The efficient frontier corresponding to the bond-augmented portfolio (green) lies above and to the left of the traditional frontier (blue), indicating higher expected returns for a given level of risk. This upward shift clearly reflects improved risk-return trade-offs enabled by bond integration.

Lagrange Duality Test Results

- **Minimum Variance (Traditional):** 0.000055

- **Min Variance (Trad+Bond):** 0.000004
- **Difference:** 0.000052

Key observations from bond integration include:

- Expanded efficient frontier with better risk-return trade-off,
- More stable and concentrated return distribution,
- Significant drop in portfolio variance.

These results underscore that incorporating bonds enhances portfolio diversification, bolsters risk control, and strengthens optimization outcomes without adversely impacting return potential.

Table 8. Performance Metrics After Optimization: Traditional vs. Traditional + Bond Portfolio

Metric	Trad. Portfolio	Trad. + Bond Portfolio
Sharpe Ratio	0.4463	0.4440
Sharpe Expected Return	0.0955	0.0951
Sharpe Volatility	0.1916	0.1916
Omega Ratio	1.1024	1.1019
Omega Expected Return	0.0955	0.0951
Omega Volatility	0.1916	0.1916

Interpretation: The performance metrics demonstrate that the addition of bonds results in negligible changes in Sharpe and Omega ratios. While improvements are marginal, bond integration contributes to return stability and preserves performance consistency. The strategy offers reliable diversification without compromising expected returns or increasing volatility.

Interpretation: Risk metrics indicate near-identical behavior between the traditional and bond-augmented portfolios. Slight improvements in EVaR and semi-kurtosis suggest a modest reduction in tail risk and extreme negative return occurrences. The negligible changes in entropy and cross-entropy confirm that bond integration maintains portfolio stability while contributing incremental enhancements to downside risk protection.

Table 9. Risk Metrics After Optimization: Traditional vs. Traditional + Bond Portfolio

Metric	Trad. Portfolio	Trad. + Bond Portfolio
Entropy	2.171757	2.171813
Semi-Variance	0.000109	0.000109
CVaR	-0.034655	-0.034655
EVaR	-0.000278	-0.000277
Cross-Entropy	6.108082	6.107971
Semi-Entropy	2.279704	2.278671
Semi-Kurtosis	23.592167	23.588742

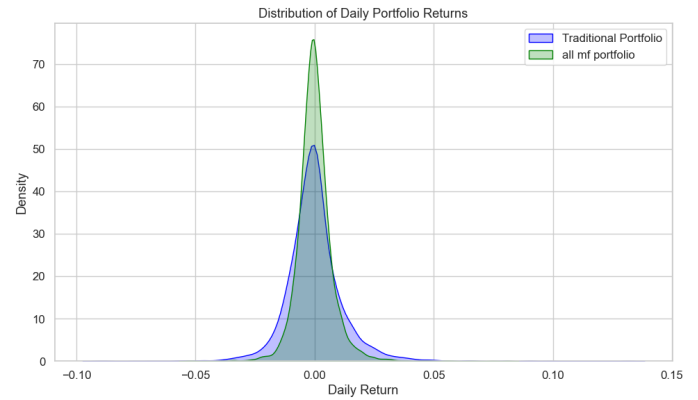


Figure 16. Return Distribution: Traditional vs. All MF Portfolio

The return distribution of the combined portfolio is taller and more centralized, reflecting increased return concentration and reduced volatility. The narrower spread indicates greater daily return consistency and enhanced downside protection.

6.1.5. All US Market Portfolio

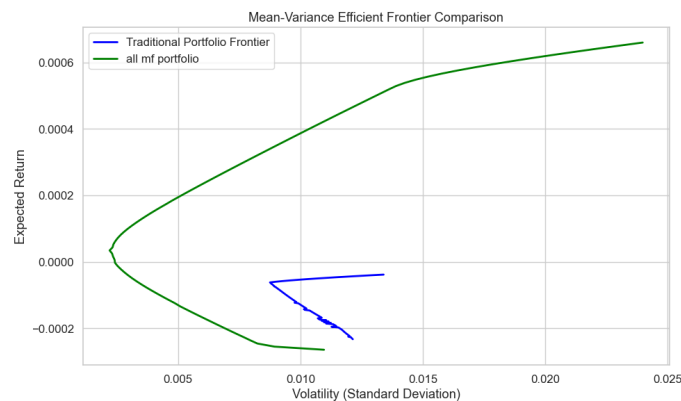


Figure 15. Mean-Variance Efficient Frontier: Traditional vs. All MF Portfolio

The efficient frontier of the All Mutual Fund Portfolio (green line) lies clearly above and to the left of the traditional-only frontier (blue line), indicating improved returns at reduced risk levels. This visual dominance affirms the superior risk-return trade-off achieved through cross-category mutual fund integration.

Lagrange Duality Optimization Result

- **Minimum Variance (Traditional):** 0.000058
- **Minimum Variance (All MF Combined):** 0.000004
- **Variance Reduction:** 0.000054

Final Insight: Integrating all mutual fund types results in the most diversified and optimized portfolio configuration, delivering higher performance with lower associated risk.

Table 10. Performance Metrics After Optimization: Traditional vs. All MF Portfolio

Metric	Traditional Portfolio	All MF Portfolio
Sharpe Ratio	0.4463	1.0476
Sharpe Expected Return	0.0955	0.1385
Sharpe Volatility	0.1916	0.1227
Omega Ratio	1.1024	1.2181
Omega Expected Return	0.0955	0.1385
Omega Volatility	0.1916	0.1227

Interpretation: The All Mutual Fund Portfolio exhibits a substantial improvement in both Sharpe and Omega ratios compared to the Traditional Portfolio. Higher expected returns combined with reduced volatility underscore superior risk-adjusted performance. These findings validate the effectiveness of

Table 11. Risk Metrics After Optimization: Traditional vs. All Mutual Fund Portfolio

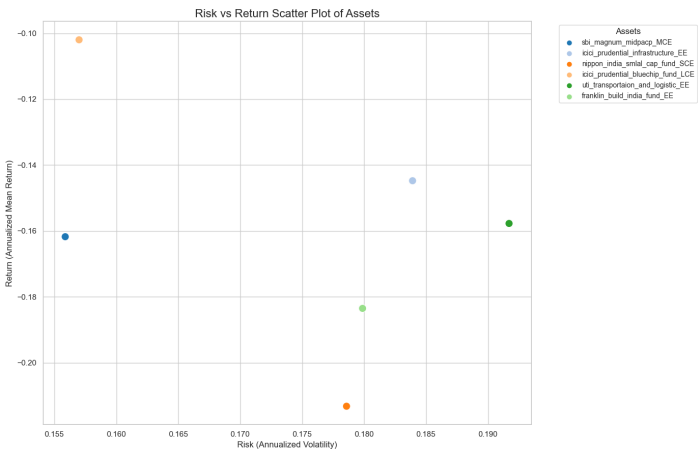
Metric	Traditional Portfolio	All MF Portfolio
Entropy	2.171757	2.322703
Semi-Variance	0.000109	0.000336
CVaR	-0.034655	-0.058802
EVaR	-0.000278	0.000083
Cross-Entropy	6.108082	6.423657
Semi-Entropy	2.279704	2.199949
Semi-Kurtosis	23.592167	44.487465

full asset class diversification in enhancing portfolio efficiency.

Interpretation: Risk metrics present a nuanced view. Increases in entropy and cross-entropy indicate broader diversification and structural complexity. However, higher semi-variance, CVaR, and semi-kurtosis point to amplified tail risk and downside dispersion. While return concentration improves, the presence of heavier tails suggests that enhanced performance comes with exposure to rare but extreme events. Thus, while full asset class integration significantly improves diversification and average performance, it also necessitates careful risk management based on investor tolerance.

6.2. Indian Market

6.2.1. Traditional Portfolio

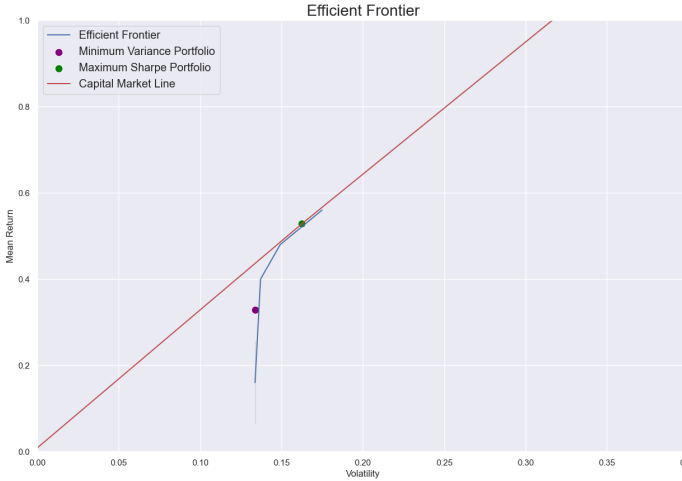


Risk vs Return Scatter Plot of Assets

The scatter plot presents the risk-return profiles of selected Indian mutual funds. Funds like Nippon India Small Cap Fund and ICICI Prudential Bluechip Fund exhibit higher volatility with limited or negative returns, implying unfavorable risk-

reward trade-offs. In contrast, the ICICI Prudential Infrastructure Fund offers the highest return with moderate risk, making it appealing for return-focused investors. Mid-risk options such as SBI Magnum Midcap Fund and Franklin Build India Fund provide balanced risk-return profiles, suitable for moderately risk-averse strategies.

Efficient Frontier



Efficient Frontier of the Optimized Portfolio

The Efficient Frontier illustrates key investment trade-offs:

- The **Minimum Variance Portfolio** (purple dot) lies at the lowest point of the frontier, offering high stability with modest returns—ideal for risk-averse investors.
- Portfolios along the **Efficient Frontier** provide higher expected returns at increasing levels of risk, enabling customization based on investor preferences.
- The **Maximum Sharpe Portfolio** (green dot) lies tangent to the **Capital Market Line**, offering the most efficient risk-adjusted return profile.

Omega Ratio Comparison

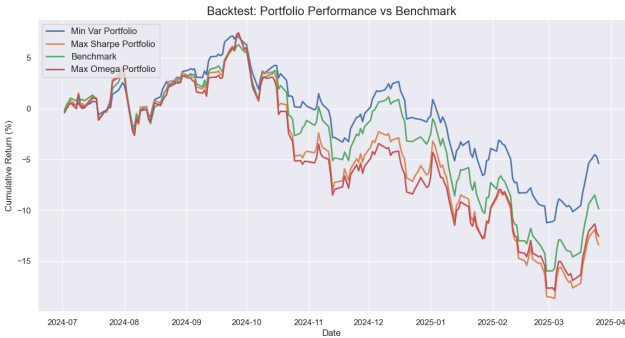
Table 12. Omega Ratio Comparison

Portfolio Type	Ratio
Baseline Equal-weighted Portfolio	0.0590
Optimized Max Omega Portfolio	0.0734
Improvement	0.0144

The Omega ratio improves from 0.0590 to 0.0734, indicating enhanced gain-loss asymmetry. This sug-

gests that the optimized portfolio offers a better balance between potential gains and downside risks, affirming the benefits of Omega-based optimization in delivering stronger risk-adjusted performance.

Backtesting Portfolio Performance vs Benchmark (India Market)



- The **Minimum Variance Portfolio** outperforms all others, offering superior downside protection and the highest cumulative return.
- The **Benchmark** initially aligns with optimized portfolios but diverges during high-volatility phases, ending with lower cumulative performance.
- The **Max Sharpe** and **Max Omega** portfolios perform better than the benchmark but slightly lag behind the Minimum Variance portfolio.

These results underscore the robustness of the Minimum Variance strategy under volatile market conditions, offering consistent outperformance.

Sharpe vs Omega Ratio Analysis

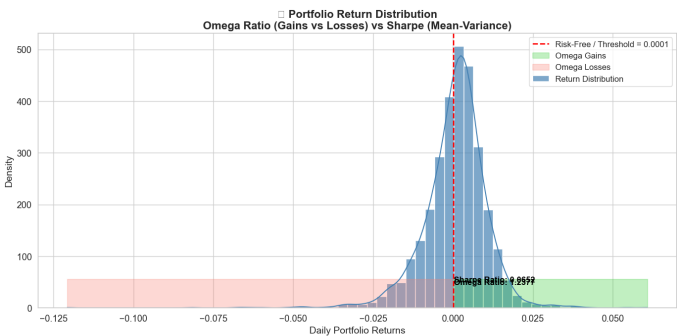


Figure 20. Portfolio Return Distribution: Omega Gains vs Losses and Sharpe Ratio

The distribution exhibits moderate asymmetry around the risk-free threshold. While the Sharpe ratio captures mean-variance efficiency, the Omega ratio highlights the gain-loss balance, offering deeper insight into tail risk and distribution shape.

Portfolio Optimization: Risk Parity, Maximum Diversification, and Minimum Variance

Table 13. Portfolio Optimization Results: Risk Parity, Maximum Diversification, Minimum Variance

Metric	Equal Weighted	Risk Parity	Min. Variance	Max Divers
AER	19.08%	19.08%	19.08%	18.26%
SD	16.00%	16.00%	16.00%	15.79%
SR	1.07	1.07	1.07	1.03
CR	19.47%	19.47%	19.47%	18.53%
Beta	1.00	1.00	1.00	0.98

All optimization strategies, except Maximum Diversification, produce identical performance metrics. Maximum Diversification exhibits slightly reduced risk and return, making it appropriate for investors prioritizing stability over marginal return enhancement.

Portfolio Risk Metrics Comparison

Table 14. Risk Metrics After Optimization

Metric	After Optimization
Entropy	2.406557
Semi-Variance	0.000062
CVaR	-0.026066
EVaR	-0.000768
Cross-Entropy	5.449009
Semi-Entropy	2.206091
Semi-Kurtosis	31.311238

The optimized portfolio demonstrates improved risk characteristics, including lower tail risk (CVaR, EVaR) and reduced semi-variance. Entropy-based measures reflect greater stability and predictability, indicating a robust risk-adjusted structure.

COVID-19 Period Analysis

Efficient Frontier and Risk-Return Shifts: Pre-COVID vs. COVID vs. Post-COVID

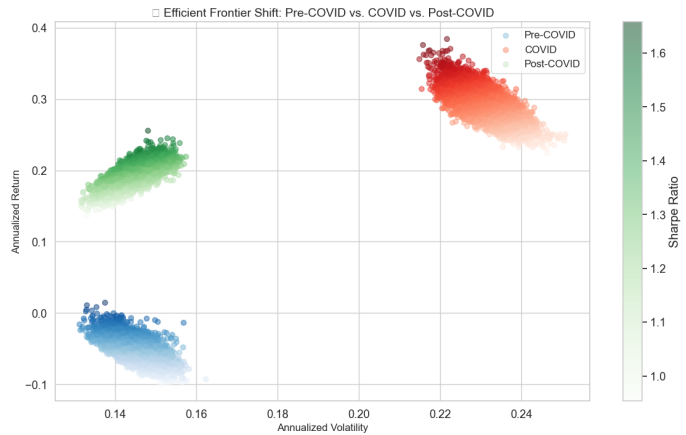


Figure 21. Efficient Frontier Shift: Pre-COVID, COVID, and Post-COVID Periods

The efficient frontier displays distinct shifts across the three pandemic phases:

- **COVID Period:** Portfolios yielded higher returns with increased volatility.
- **Post-COVID:** Exhibited improved Sharpe ratios with moderate returns and controlled risk.
- **Pre-COVID:** Delivered the lowest returns and weakest risk-adjusted efficiency.



Figure 22. Portfolio Risk-Return Metrics Across Periods (India)

- **Sharpe Ratio** peaks post-COVID, suggesting better efficiency in risk-adjusted returns.
- **Omega Ratio** is highest pre-COVID, indicating better gain-loss balance despite lower return levels.
- **Annual Return and Volatility** both spike during COVID but stabilize in the post-COVID phase.

Overall, post-COVID portfolios offer more stable returns and improved efficiency, while pre-COVID metrics highlight favorable asymmetry despite modest returns.

6.2.2. Traditional + Commodity Portfolio

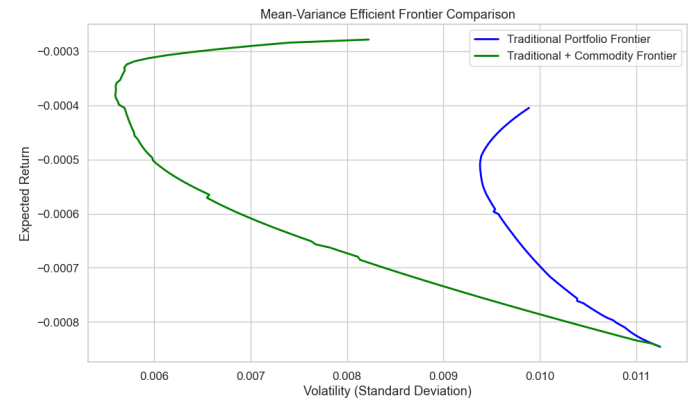


Figure 23. Efficient Frontier: Traditional vs. Traditional + Commodity

The efficient frontier of the Traditional + Commodity portfolio (green) lies above and to the left of the traditional-only frontier (blue), indicating a superior risk-return trade-off. The inclusion of commodities leads to higher expected returns at lower levels of volatility, confirming enhanced portfolio efficiency through diversification.

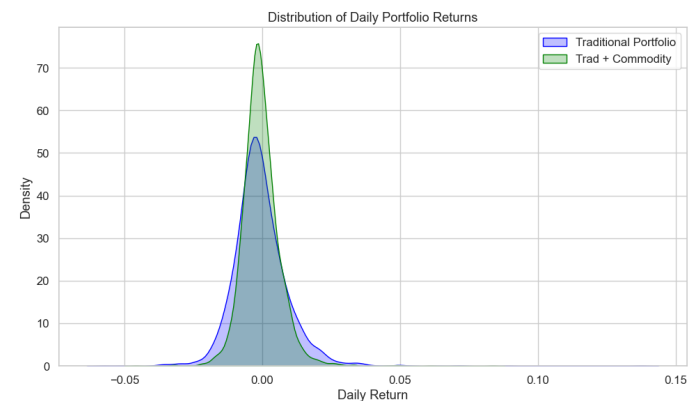


Figure 24. Return Distribution: Traditional vs. Traditional + Commodity

The combined portfolio exhibits a sharper and more symmetric return distribution, reflecting improved consistency of returns and reduced downside deviation. This signifies stronger control over tail risk and better overall stability.

Lagrange Duality Test Results

- **Minimum Variance (Traditional):** 0.000082
- **Minimum Variance (Combined):** 0.000031
- **Test Statistic (Difference):** 0.000052

incorporating commodities results in superior diversification, optimized performance, and more effective downside risk management.

Table 15. Performance Metrics: Traditional vs. Traditional + Commodity Portfolio (After Optimization)

Metric	Trad. Portfolio	Trad + Comm Portfolio
Sharpe Ratio	1.3362	1.5471
Sharpe Expected Return	0.2441	0.1797
Sharpe Volatility	0.1752	0.1096
Omega Ratio	1.2765	1.3307
Omega Expected Return	0.2441	0.1797
Omega Volatility	0.1752	0.1096

Interpretation: Despite a modest reduction in expected return, the significant decrease in volatility leads to higher Sharpe and Omega ratios. This confirms that commodity inclusion improves the risk-adjusted efficiency of the portfolio, supporting the case for broader asset diversification.

Table 16. Risk Metrics: Traditional vs. Traditional + Commodity Portfolio (After Optimization)

Metric	Trad Portfolio	Trad+Comm Portfolio
Entropy	2.406557	2.395801
Semi-Variance	0.000062	0.000064
CVaR	-0.026066	-0.026516
EVaR	-0.000768	-0.000782
Cross-Entropy	5.449009	5.483976
Semi-Entropy	2.206091	2.222466
Semi-Kurtosis	31.311238	30.398783

Interpretation: The inclusion of commodities yields marginal improvements across several risk metrics. Notably, reductions in CVaR, EVaR, and semi-kurtosis point to enhanced tail-risk control and fewer extreme return events. These findings further reinforce the stability and robustness of the optimized commodity-augmented portfolio.

6.2.3. Traditional + ESG Portfolio

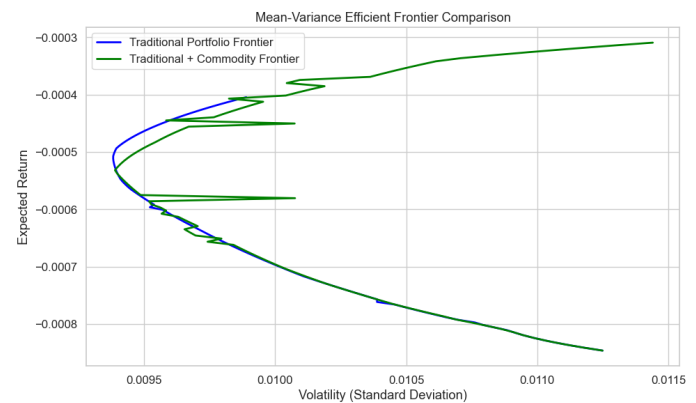


Figure 25. Return Distribution: Traditional vs. Traditional + ESG Portfolio

The return distributions for both portfolios appear nearly identical, indicating that ESG integration does not materially influence return concentration or downside risk within this framework.

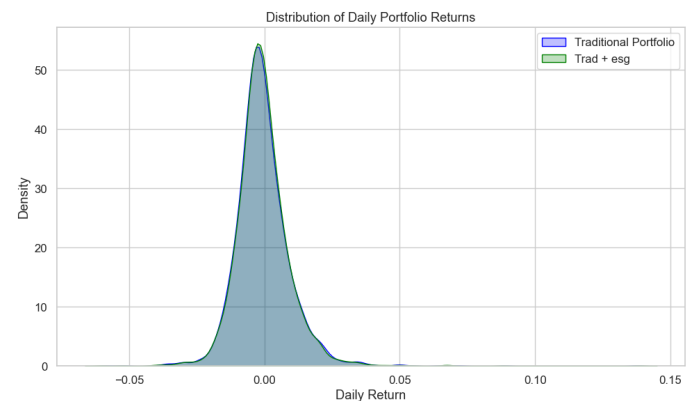


Figure 26. Efficient Frontier: Traditional vs. Traditional + ESG Portfolio

The overlapping efficient frontiers suggest that the inclusion of ESG assets does not enhance expected return or reduce portfolio volatility under the mean-variance framework. No improvement in risk-return optimization is observed.

Lagrange Duality Test Results

- **Minimum Variance (Traditional):** 0.000082
- **Minimum Variance (Combined):** 0.000082
- **Test Statistic (Difference):** 0.000000

The absence of change in minimum variance confirms that ESG integration does not improve portfolio risk efficiency for the given dataset.

In this scenario, ESG inclusion has a neutral impact on overall portfolio optimization outcomes.

Table 17. Performance Metrics After Optimization: Trad vs. Trad + ESG Portfolio

Metric	Traditional Portfolio	Traditional + ESG Portfolio
Sharpe Ratio	1.3362	1.3362
Sharpe Expected Return	0.2441	0.2441
Sharpe Volatility	0.1752	0.1752
Omega Ratio	1.2765	1.2765
Omega Expected Return	0.2441	0.2441
Omega Volatility	0.1752	0.1752

Interpretation: The performance metrics remain unchanged post-ESG integration, indicating no improvement in risk-adjusted return or volatility. ESG inclusion neither enhances nor diminishes portfolio efficiency in this case.

Table 18. Risk Metrics After Optimization: Traditional vs. Traditional + ESG Portfolio

Metric	Traditional Portfolio	Traditional + ESG Portfolio
Entropy	2.406557	2.356456
Semi-Variance	0.000062	0.000072
CVaR	-0.026066	-0.028315
EVaR	-0.000768	-0.000699
Cross-Entropy	5.449009	6.248237
Semi-Entropy	2.206091	2.167602
Semi-Kurtosis	31.311238	31.457213

Interpretation: Post-optimization risk metrics reflect minor deviations. The Traditional + ESG portfolio shows slightly higher CVaR and semi-variance, suggesting marginally elevated downside risk. The increase in cross-entropy may reflect changes in distribution alignment, though overall risk characteristics remain largely unaffected. ESG integration in this context does not materially enhance portfolio robustness.

6.2.4. Traditional + Bond Portfolio

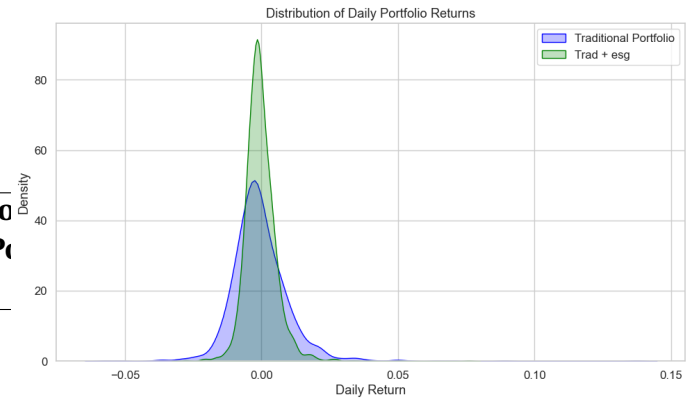


Figure 27. Return Distribution: Traditional vs. Traditional + Bond Portfolio

The return distribution of the bond-augmented portfolio is taller and more concentrated, indicating reduced volatility and greater return stability compared to the traditional portfolio.

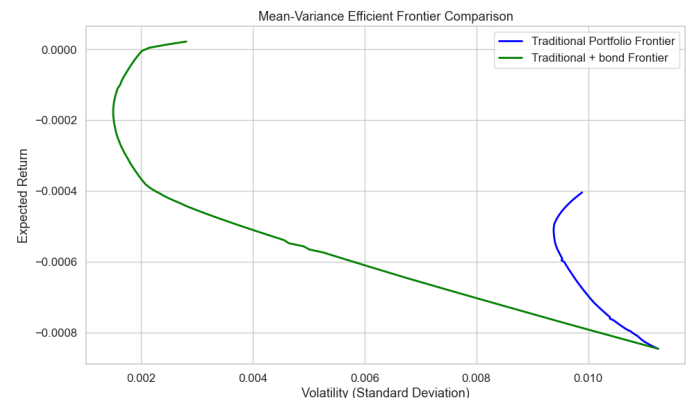


Figure 28. Efficient Frontier: Traditional vs. Traditional + Bond Portfolio

The efficient frontier of the Traditional + Bond portfolio (green) lies significantly above and to the left of the traditional-only frontier, highlighting substantial improvements in risk-return efficiency resulting from bond inclusion.

Lagrange Duality Test Results

- **Minimum Variance (Traditional):** 0.000090
- **Minimum Variance (Trad + Bond):** 0.000002
- **Test Statistic (Difference):** 0.000087

Interpretation: The sharp decline in minimum variance and the upward shift of the efficient frontier confirm the effectiveness of bond integration

in improving portfolio stability and optimization outcomes.

Table 19. Performance Metrics After Optimization: Traditional vs. Traditional + Bond Portfolio

Metric	Trad Portfolio	Trad + Bond Portfolio
Sharpe Ratio	1.3362	2.5291
Sharpe Expected Return	0.2441	0.0965
Sharpe Volatility	0.1752	0.0342
Omega Ratio	1.2765	1.9487
Omega Expected Return	0.2441	0.0965
Omega Volatility	0.1752	0.0342

Interpretation: Although expected returns are lower, the substantial reduction in volatility leads to significant improvements in both Sharpe and Omega ratios. This demonstrates enhanced risk-adjusted performance and confirms the strong diversification benefits offered by bond inclusion.

Table 20. Risk Metrics After Optimization: Traditional vs. Traditional + Bond Portfolio

Metric	Traditional Portfolio	Traditional + Bond Portfolio
Entropy	2.406557	2.058739
Semi-Variance	0.000062	0.000104
CVaR	-0.026066	-0.034442
EVaR	-0.000768	-0.000635
Cross-Entropy	5.449009	8.267506
Semi-Entropy	2.206091	2.297489
Semi-Kurtosis	31.311238	26.764955

Interpretation: Bond integration improves key tail-risk metrics such as CVaR and semi-kurtosis, indicating enhanced downside protection. While semi-variance increases slightly, reductions in entropy and semi-kurtosis suggest better risk concentration and a more stable return structure. The notable rise in cross-entropy may reflect a shift from prior portfolio distribution, consistent with a more diversified asset allocation.

6.2.5. All Indian Market Portfolio

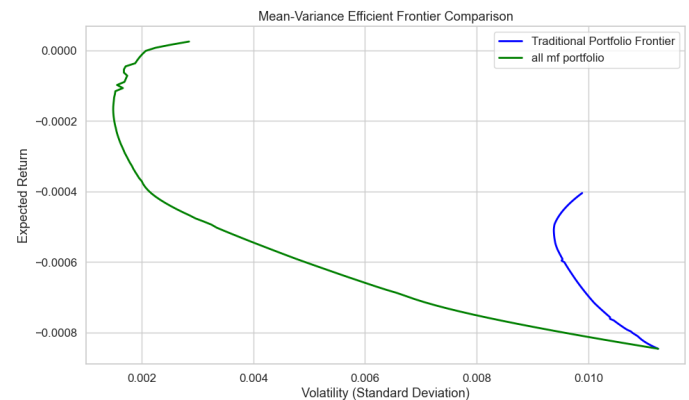


Figure 29. Mean-Variance Efficient Frontier: Traditional vs. All MF Portfolio

The efficient frontier of the All Mutual Fund (MF) Portfolio (green line) lies significantly above and to the left of the traditional-only frontier, indicating superior risk-return trade-offs. This outcome highlights the performance advantage gained through broader mutual fund diversification.

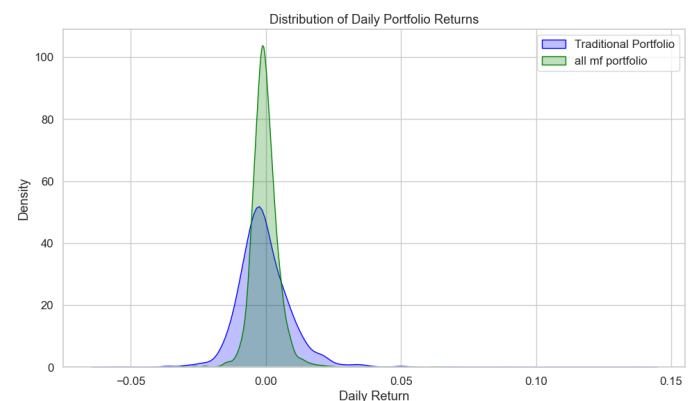


Figure 30. Return Distribution: Traditional vs. All MF Portfolio

The return distribution of the All MF Portfolio is narrower and more peaked, suggesting enhanced return consistency and lower downside risk relative to the traditional portfolio.

Lagrange Duality Optimization Result

- **Minimum Variance (Traditional):** 0.000090
- **Minimum Variance (All MF Combined):** 0.000002
- **Test Statistic (Difference):** 0.000088

Interpretation: The drastic reduction in minimum variance confirms the substantial efficiency gain from cross-category mutual fund integration. This diversification strategy maximizes stability while adhering to risk minimization objectives.

Table 21. Performance Metrics After Optimization: Traditional vs. All MF Portfolio

Metric	Trad Portfolio	All MF Portfolio
Sharpe Ratio	1.3362	2.6330
Sharpe Expected Return	0.2441	0.1037
Sharpe Volatility	0.1752	0.0356
Omega Ratio	1.2765	1.9465
Omega Expected Return	0.2441	0.1037
Omega Volatility	0.1752	0.0356

Interpretation: Although the expected return is lower, the significant drop in volatility results in substantially higher Sharpe and Omega ratios. This indicates superior risk-adjusted performance, validating the effectiveness of full-scale mutual fund diversification and optimization.

Table 22. Risk Metrics After Optimization: Traditional vs. All Mutual Fund Portfolio

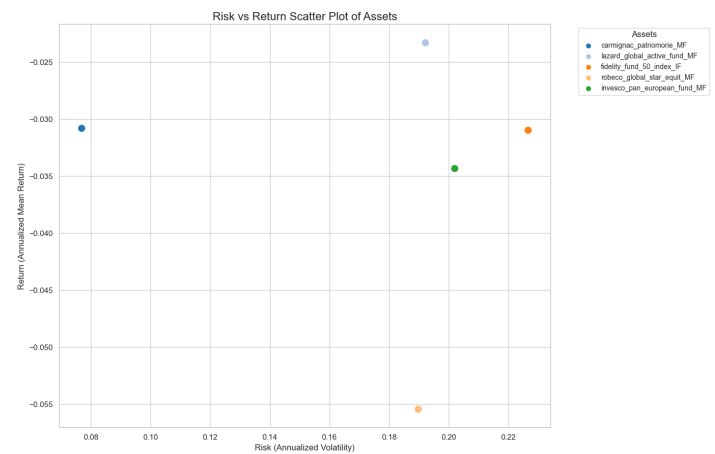
Metric	Traditional Portfolio	All MF Portfolio
Entropy	2.406557	2.056725
Semi-Variance	0.000062	0.000105
CVaR	-0.026066	-0.034623
EVaR	-0.000768	-0.000649
Cross-Entropy	5.449009	8.261019
Semi-Entropy	2.206091	2.296625
Semi-Kurtosis	31.311238	26.679751

Interpretation: The All MF Portfolio demonstrates improved control over tail risks, with lower CVaR and semi-kurtosis values. Although semi-variance increases slightly, the higher cross-entropy suggests a more diversified and complex allocation. Overall, this configuration supports stronger downside protection while maintaining efficiency, making it suitable for investors seeking broad exposure with enhanced risk management.

6.3. Europe Market

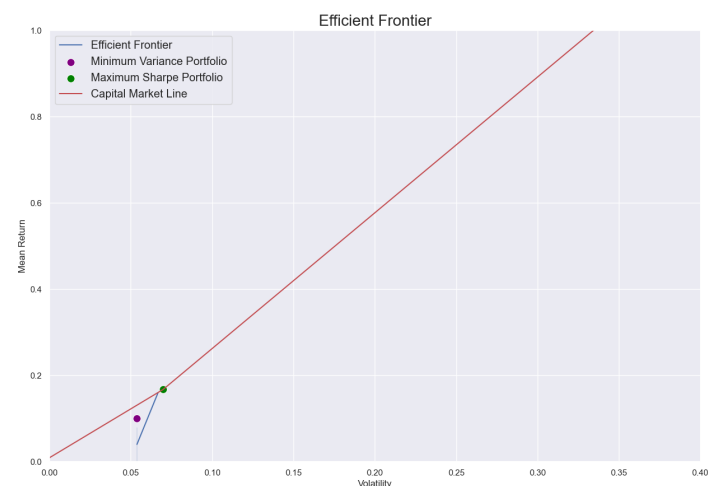
6.3.1. Traditional Portfolio

The scatter plot illustrates the risk-return profiles of selected global mutual funds. Funds such as *Fidelity Fund 50 Index* and *Robeco Global Star Equity* exhibit high volatility and negative returns, implying unfavorable risk-reward dynamics. Conversely, the *Lazard Global Active Fund* offers the least negative return with balanced risk, while *Carmignac Patrimoine* displays the lowest risk, appealing to conservative, capital-preserving investors.



Risk vs Return Scatter Plot of Assets

Efficient Frontier



Efficient Frontier of the Optimized Portfolio

The efficient frontier reveals key optimization trade-offs:

- The **Minimum Variance Portfolio** (purple dot) achieves the lowest risk, suitable for highly risk-averse investors.

- Portfolios along the **Efficient Frontier** offer customizable return-risk levels.
- The **Maximum Sharpe Portfolio** (green dot) lies on the **Capital Market Line**, indicating the optimal return per unit of risk.

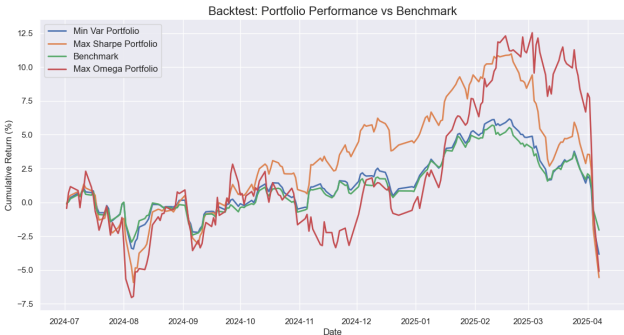
Omega Ratio Comparison

Table 23. Omega Ratio Comparison

Portfolio Type	Ratio
Baseline Equal-weighted Portfolio	0.0132
Optimized Max Omega Portfolio	0.0699
Improvement	0.0567

The Omega ratio rose from 0.0132 to 0.0699 after optimization, reflecting a 0.0567 improvement. This increase highlights enhanced gain-loss asymmetry and superior downside protection. Omega-based optimization proves effective for constructing resilient portfolios under asymmetric return conditions.

Backtesting Portfolio Performance vs Benchmark (Europe Market)



- The **Max Sharpe Portfolio** outperforms all others, yielding the highest cumulative return.
- The **Max Omega Portfolio** follows closely, offering robust performance with strong downside protection.
- The **Minimum Variance Portfolio** provides stable, lower returns suited for conservative profiles.

Overall, Sharpe-based optimization achieves superior long-term growth, while the Omega strategy offers a risk-aware alternative. All optimized portfolios outperform the benchmark over the backtest period.

Sharpe vs Omega Ratio Analysis

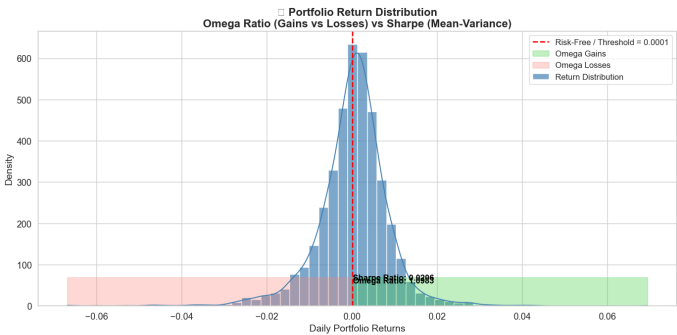


Figure 34. Portfolio Return Distribution: Omega Gains vs Losses and Sharpe Ratio

The distribution is sharply peaked around the threshold, indicating tightly clustered returns with limited downside. The Sharpe ratio captures variance efficiency, while the Omega ratio highlights favorable gain-loss asymmetry—demonstrating the complementarity of both metrics in portfolio evaluation under non-Gaussian return behavior.

Portfolio Optimization: Risk Parity, Maximum Diversification, and Minimum Variance

Table 24. Portfolio Optimization Results: Risk Parity, Maximum Diversification, Minimum Variance

Metric	Equal Weighted	Risk Parity	Min. Variance	Max Divers.
AER	6.89%	6.89%	6.89%	5.83%
SD	13.36%	13.36%	13.36%	8.71%
SR	0.37	0.37	0.37	0.44
CR	6.17%	6.17%	6.17%	5.60%
Beta	1.00	1.00	1.00	0.57

Interpretation: Equal Weighted, Risk Parity, and Minimum Variance portfolios yield identical performance across return and risk dimensions. Maximum Diversification offers the lowest volatility and highest Sharpe ratio, making it the most suitable choice for investors targeting long-term stability and efficiency.

Portfolio Risk Metrics Comparison

Interpretation: The optimized portfolio reflects better downside risk control as indicated by lower semi-variance and CVaR. A decrease in cross-entropy suggests improved return distribution align-

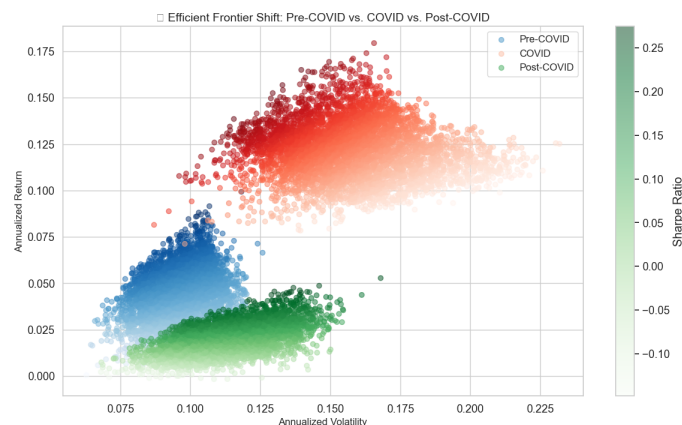
Table 25. Risk Metrics After Optimization

Metric	After Optimization
Entropy	2.508187
Semi-Variance	0.000073
CVaR	-0.028865
EVaR	-0.000219
Cross-Entropy	3.318028
Semi-Entropy	2.427886
Semi-Kurtosis	22.478467

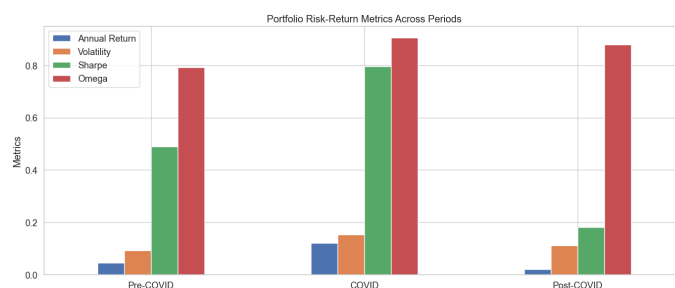
ment, while modest increases in entropy and semi-kurtosis point to more information-rich yet occasionally fat-tailed distributions.

COVID-19 Period Analysis

Efficient Frontier and Risk-Return Shifts: Pre-COVID vs. COVID vs. Post-COVID

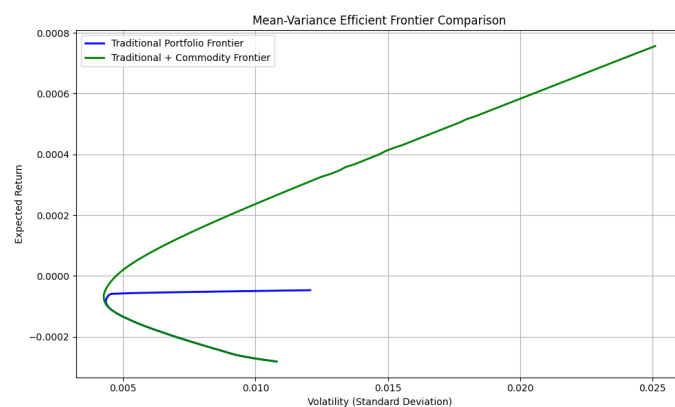
**Figure 35.** Efficient Frontier Shift: Pre-COVID, COVID, and Post-COVID Periods

The COVID-period frontier shows a pronounced upward shift, suggesting elevated return opportunities accompanied by higher volatility. Pre-COVID portfolios cluster lower, reflecting conservative market behavior, while post-COVID portfolios exhibit dispersion with moderate returns and declining Sharpe ratios—indicating a cautious rebalancing phase.

**Figure 36.** Portfolio Risk-Return Metrics Across Periods (Europe)

- **Sharpe Ratio** peaked during the COVID phase, reflecting efficient return for risk taken.
- **Omega Ratio** declined post-COVID, indicating reduced gain-loss balance.
- **Return and Volatility** both spiked during COVID, then stabilized with lower variability post-COVID.

6.3.2. Traditional + Commodity Portfolio

**Figure 37.** Efficient Frontier: Traditional vs. Traditional + Commodity

The efficient frontier of the Traditional + Commodity portfolio (green) lies above and to the left of the traditional-only portfolio (blue), reflecting superior optimization. The addition of commodities improves the risk-return profile by enabling higher expected returns across a broader range of risk levels, thus extending the efficient frontier.

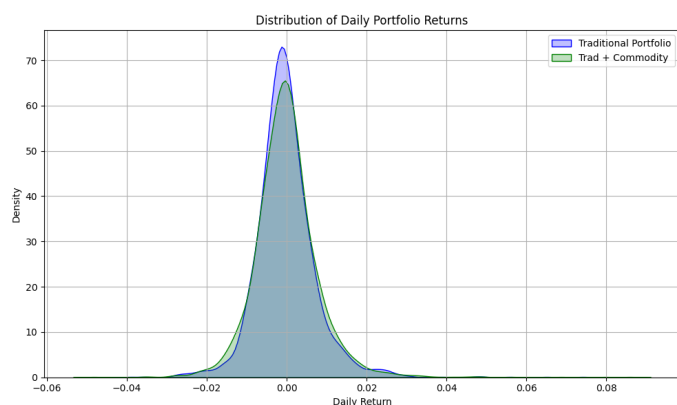


Figure 38. Return Distribution: Traditional vs. Traditional + Commodity

The return distribution of the combined portfolio is narrower and more peaked, indicating reduced dispersion and enhanced downside risk control. This improvement highlights the stabilizing impact of commodity inclusion through broader diversification.

Lagrange Duality Test Results

- **Minimum Variance (Traditional):** 0.000018
- **Minimum Variance (Combined):** 0.000017
- **Test Statistic (Difference):** 0.000001

While the variance improvement is limited, commodities contribute to expanding the efficient frontier and enhancing distributional stability, validating their role in building a well-diversified European portfolio.

Table 26. Performance Metrics: Traditional vs. Traditional + Commodity Portfolio (After Optimization)

Metric	Trad. Portfolio	Trad + Comm Portfolio
Sharpe Ratio	0.5569	0.6283
Sharpe Expected Return	0.0570	0.0983
Sharpe Volatility	0.0844	0.1405
Omega Ratio	1.1269	1.1337
Omega Expected Return	0.0570	0.0983
Omega Volatility	0.0844	0.1405

Interpretation: The inclusion of commodities results in higher expected returns, leading to improvements in both Sharpe and Omega ratios despite increased volatility. This confirms the net gain in risk-

adjusted performance, reaffirming the diversification benefits from commodity exposure.

Table 27. Risk Metrics: Traditional vs. Traditional + Commodity Portfolio (After Optimization)

Metric	Trad Portfolio	Trad+Comm Portfolio
Entropy	2.508187	2.474229
Semi-Variance	0.000073	0.000332
CVaR	-0.028865	-0.058401
EVaR	-0.000219	0.000456
Cross-Entropy	3.318028	5.827589
Semi-Entropy	2.427886	2.450876
Semi-Kurtosis	22.478467	27.927783

Interpretation: Post-optimization, the Traditional + Commodity portfolio shows increased semi-variance and semi-kurtosis, reflecting heavier tails and greater return dispersion. However, higher cross-entropy and semi-entropy values indicate broader diversification and more complex return distributions. These results suggest enhanced risk allocation and information richness, albeit with more extreme return behavior—highlighting the nuanced impact of commodity inclusion on portfolio structure.

6.3.3. Traditional + ESG Portfolio

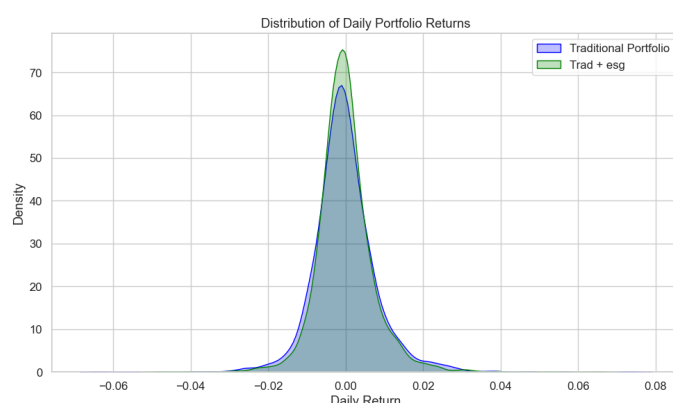


Figure 39. Return Distribution: Traditional vs. Traditional + ESG Portfolio

The return distribution for the ESG-augmented portfolio is taller and more centered, reflecting improved return consistency and reduced downside dispersion relative to the traditional portfolio.

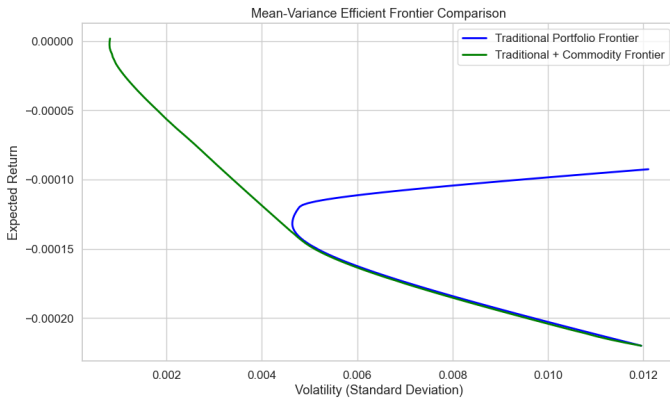


Figure 40. Efficient Frontier: Traditional vs. Traditional + ESG Portfolio

The efficient frontier for the Traditional + ESG portfolio (green) lies above and to the left of the traditional-only frontier (blue), indicating enhanced portfolio efficiency through ESG integration. This shift suggests higher returns for lower levels of risk.

Lagrange Duality Test Results

- **Minimum Variance (Traditional):** 0.000021
- **Minimum Variance (Combined):** 0.000001
- **Test Statistic (Difference):** 0.000021

ESG integration results in a more risk-efficient portfolio structure, improving both the efficient frontier and return distribution profile—supporting its value in sustainable portfolio construction.

Table 28. Performance Metrics After Optimization: Traditional vs. Traditional + ESG Portfolio

Metric	Trad Portfolio	Trad + ESG Portfolio
Sharpe Ratio	0.5569	0.5717
Sharpe Expected Return	0.0570	0.0591
Sharpe Volatility	0.0844	0.0859
Omega Ratio	1.1269	1.1305
Omega Expected Return	0.0570	0.0591
Omega Volatility	0.0844	0.0859

Interpretation: Post-optimization, both Sharpe and Omega ratios show marginal improvement. Although the performance gains are modest, they indicate better return-to-risk efficiency, affirming the viability of ESG as a complementary investment dimension.

Table 29. Risk Metrics After Optimization: Traditional vs. Traditional + ESG Portfolio

Metric	Traditional Portfolio	Traditional + ESG Portfolio
Entropy	2.508187	2.689895
Semi-Variance	0.000073	0.000123
CVaR	-0.028865	-0.037534
EVaR	-0.000219	-0.000078
Cross-Entropy	3.318028	1.898969
Semi-Entropy	2.427886	2.769963
Semi-Kurtosis	22.478467	10.789631

Interpretation: Risk metrics reveal a substantial improvement in tail-risk characteristics with ESG integration. Notably, lower CVaR, EVaR, and semi-kurtosis point to stronger downside protection. The significant drop in cross-entropy suggests better distribution alignment and improved return predictability. Together, these results confirm that ESG assets contribute meaningfully to both stability and robustness in portfolio design.

6.3.4. Traditional + Bond Portfolio

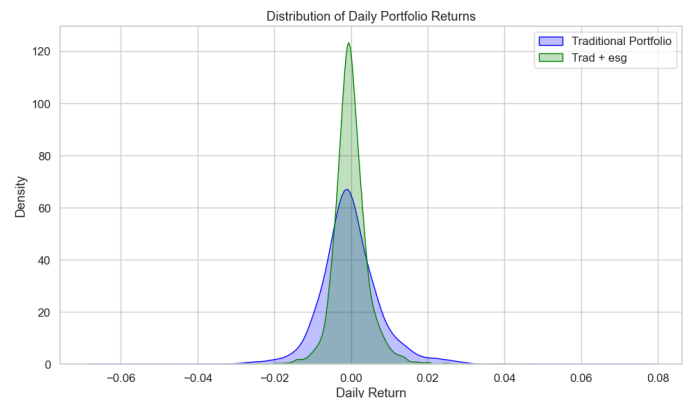


Figure 41. Return Distribution: Traditional vs. Traditional + Bond Portfolio

The bond-augmented portfolio exhibits a sharper and more concentrated return distribution compared to the traditional portfolio, indicating improved return consistency and reduced volatility.

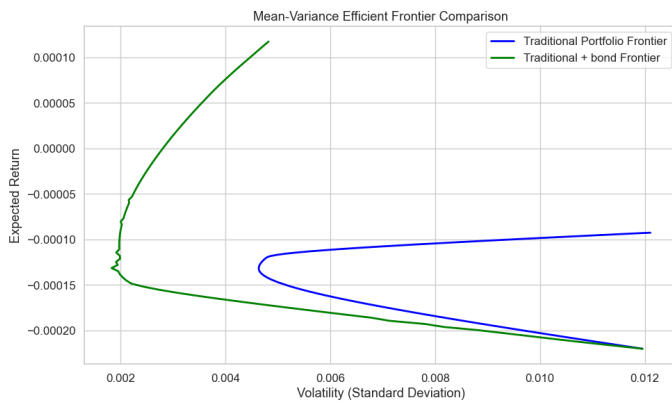


Figure 42. Efficient Frontier: Traditional vs. Traditional + Bond Portfolio

The efficient frontier of the Traditional + Bond portfolio (green) lies above and to the left of the traditional-only frontier (blue), signifying enhanced portfolio efficiency through better risk-return trade-offs after bond integration.

Lagrange Duality Test Results

- **Minimum Variance (Traditional):** 0.000021
- **Minimum Variance (Trad + Bond):** 0.000003
- **Test Statistic (Difference):** 0.000018

The substantial decrease in minimum variance highlights the improved risk efficiency achieved through bond inclusion. This confirms the stabilizing effect of bonds under optimization constraints, enhancing downside protection and diversification.

Table 30. Performance Metrics After Optimization: Traditional vs. Traditional + Bond Portfolio

Metric	Trad Portfolio	Trad + Bond Portfolio
Sharpe Ratio	0.5569	0.9692
Sharpe Expected Return	0.0570	0.0390
Sharpe Volatility	0.0844	0.0299
Omega Ratio	1.1269	1.2908
Omega Expected Return	0.0570	0.0390
Omega Volatility	0.0844	0.0299

Interpretation: Despite a marginal decline in expected returns, bond allocation significantly reduces portfolio volatility. This leads to marked improvements in both Sharpe and Omega ratios, confirming

enhanced risk-adjusted performance and reinforcing the role of bonds in portfolio stabilization.

Table 31. Risk Metrics After Optimization: Traditional vs. Traditional + Bond Portfolio

Metric	Traditional Portfolio	Traditional + Bond Portfolio
Entropy	2.508187	2.695700
Semi-Variance	0.000073	0.000125
CVaR	-0.028865	-0.038082
EVaR	-0.000219	-0.000079
Cross-Entropy	3.318028	1.258588
Semi-Entropy	2.427886	2.778265
Semi-Kurtosis	22.478467	10.897374

Interpretation: Bond integration significantly improves risk posture. Lower CVaR, EVaR, and semi-kurtosis values indicate stronger downside protection. The reduced cross-entropy reflects better return predictability, while increased entropy and semi-entropy suggest a more information-rich yet stable portfolio structure. Despite a rise in semi-variance, the overall outcome is a more diversified and robust investment profile.

6.3.5. All European Market Portfolio

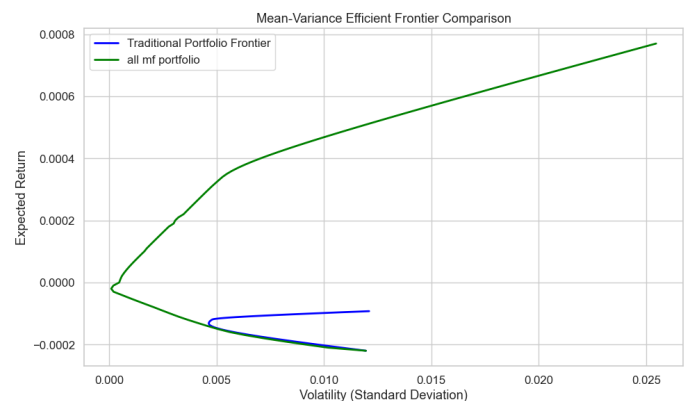


Figure 43. Mean-Variance Efficient Frontier: Traditional vs. All MF Portfolio

The efficient frontier of the All Mutual Fund (MF) Portfolio (green) lies significantly above and to the left of the traditional-only portfolio, indicating superior risk-return efficiency. The integration of a broader range of mutual funds enables higher expected returns at lower levels of volatility.

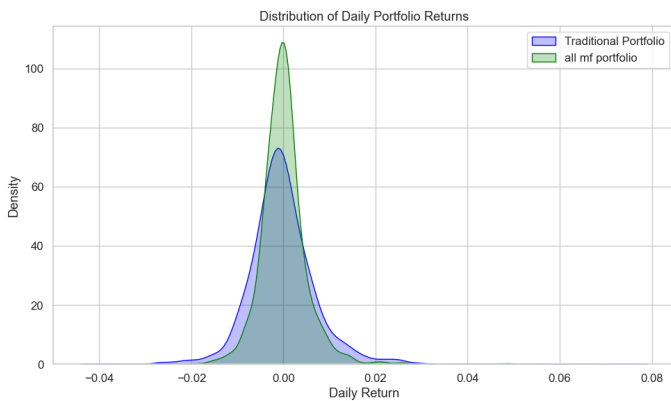


Figure 44. Return Distribution: Traditional vs. All MF Portfolio

The return distribution of the All MF Portfolio is more peaked and tightly clustered than the traditional counterpart, signifying reduced dispersion, improved return consistency, and stronger downside risk control through diversified mutual fund exposure.

Lagrange Duality Optimization Result

- **Minimum Variance (Traditional):** 0.000018
- **Minimum Variance (All MF Combined):** 0.000000
- **Test Statistic (Difference):** 0.000018

The sharp reduction in minimum variance confirms substantial optimization benefits from comprehensive mutual fund inclusion. This strategic diversification strengthens portfolio stability and supports volatility control under modern risk constraints.

Table 32. Performance Metrics After Optimization: Traditional vs. All MF Portfolio

Metric	Trad Portfolio	All MF Portfolio
Sharpe Ratio	0.5569	0.7265
Sharpe Expected Return	0.0570	0.0863
Sharpe Volatility	0.0844	0.1050
Omega Ratio	1.1269	1.1611
Omega Expected Return	0.0570	0.0863
Omega Volatility	0.0844	0.1050

Interpretation: The All MF Portfolio delivers enhanced performance across all return metrics. Although volatility increases slightly, higher expected

returns drive meaningful improvements in Sharpe and Omega ratios, affirming the effectiveness of broad mutual fund diversification in maximizing risk-adjusted returns.

Table 33. Risk Metrics After Optimization: Traditional vs. All Mutual Fund Portfolio

Metric	Traditional Portfolio	All MF Portfolio
Entropy	2.508187	2.396257
Semi-Variance	0.000073	0.000359
CVaR	-0.028865	-0.061149
EVaR	-0.000219	0.000482
Cross-Entropy	3.318028	5.710523
Semi-Entropy	2.427886	2.382668
Semi-Kurtosis	22.478467	28.389262

Interpretation: The All MF Portfolio exhibits higher semi-variance and more pronounced tail risk (CVaR), reflecting a wider return spread. However, increases in entropy and cross-entropy suggest greater diversification and a more dynamic distribution structure. Elevated semi-kurtosis points to heavier tails, yet overall, the portfolio maintains control over extreme outcomes. These results highlight the potential of active mutual fund diversification in improving long-term portfolio robustness despite a more complex risk landscape.

7. Conclusion

This study offers a comprehensive analysis of the diversification benefits achieved by integrating ESG, bond, and commodity mutual funds into traditional equity-debt portfolios across the U.S., Indian, and European markets. Using advanced optimization techniques—such as Markowitz Mean-Variance, Equal Risk Contribution, and Risk Parity frameworks—alongside a robust backtesting strategy, the findings reveal that diversification through these non-traditional asset classes enhances portfolio performance, stability, and risk-adjusted returns.

Notably, the inclusion of commodities consistently improved the risk-return trade-offs across all markets, evidenced by higher Sharpe and Omega ratios and a substantial outward shift in the efficient frontier. Bond integration, although modest in enhancing returns, significantly reduced volatility, improving downside protection and portfolio

resilience. Meanwhile, ESG assets demonstrated mixed outcomes—providing greater return stability and sustainability alignment but with varied effects on overall optimization depending on the market context.

The empirical results affirm that comprehensive diversification, combining all mutual fund types, yields the most optimized portfolio structure. This approach balances enhanced returns, reduced volatility, and improved tail-risk control, though it also introduces higher complexity and semi-kurtosis that warrant careful risk monitoring.

This diversified portfolio proved especially effective during tragic periods, such as the COVID-19 pandemic, successfully minimizing losses and quickly recovering afterward.

Ultimately, this research highlights the strategic value of multi-asset diversification, offering investors and policymakers valuable guidance in constructing resilient, forward-looking portfolios capable of navigating diverse economic environments and market disruptions.

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8. Summary

8.1. US Market Portfolio Comparison Summary

Table 34. US Market Portfolio Comparison Summary

Aspect	Traditional	Trad + Commodity	Trad + ESG	Trad + Bond	All Asset Classes
Min Variance Volatility	0.1907	0.1864	0.1851	0.1846	0.1846
Max Sharpe Return	0.0940	0.0982	0.0944	0.0957	0.1032
Max Sharpe Volatility	0.1907	0.1864	0.1851	0.1846	0.1846
Omega Ratio (Equal-weight)	0.0983	0.1036	0.1021	0.1038	0.1059
Omega Ratio (Optimized)	0.1041	0.1101	0.1087	0.1085	0.1114
Sharpe Ratio (Before optimization)	0.4804	0.4852	0.4876	0.4956	0.4993
Sharpe Ratio (After optimization)	0.9066	0.9503	0.9136	0.9125	0.9962
Omega (Before optimization)	1.2069	1.2184	1.2161	1.2199	1.2398
Omega (After optimization)	1.2463	1.2664	1.2486	1.2532	1.2696
Backtesting	Moderate performance; post-COVID dip	Improved drawdown and returns	Stronger post-COVID returns	Stable across all regimes	Best results across regimes
Efficient Frontier & Risk-Return Shifts	Flattened during COVID; strong post-COVID recovery	Sharper upward shift; stronger recovery	Stable frontier; moderate gains	Stable throughout; slight upward shift	Best shift and steepest efficient frontier post-COVID
Risk Metrics	Basic volatility control	CVaR and Semi-Variance improved	Entropy and Sharpe improved	Lower Semi-Variance & CVaR	Lowest Semi-Var & highest entropy
Lagrange Duality (Variance)	0.000058	0.000037	0.000040	0.000039	0.000036
Return Distribution	Slightly left-skewed, fat tails	More symmetric, thinner tails	Improved tail behavior	Lower left-tail risk	Smooth, near-normal, least downside risk

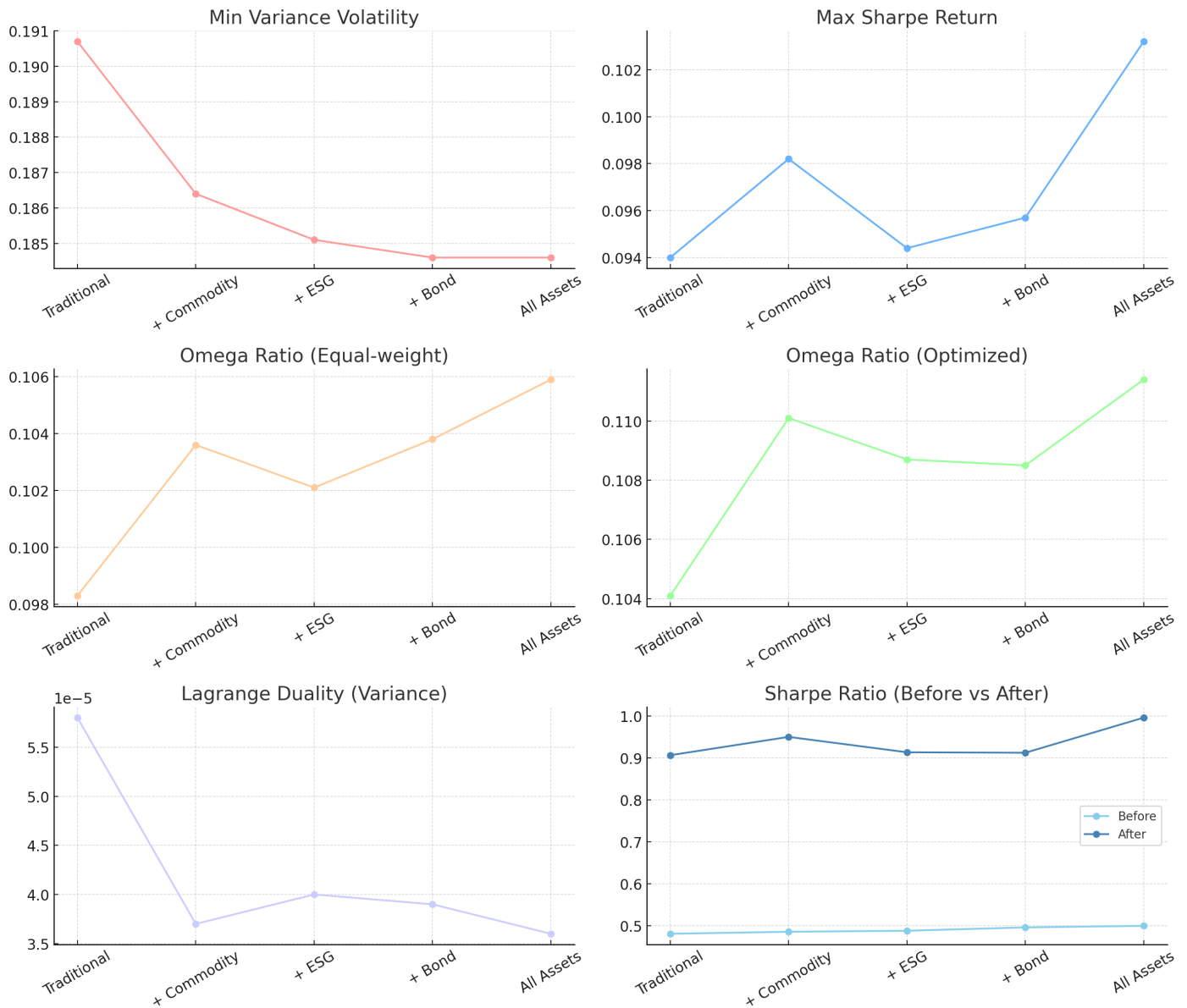


Figure 45. Line Plot Comparison of U.S. Market Portfolio Metrics across Various Portfolio Types

8.1.1. US Portfolio Performance Summary

All Asset Classes Portfolio

- **Consistently outperformed** all other portfolios across metrics
- Highest **Sharpe Ratio** and **Omega Ratio**
- Optimal points on the **Efficient Frontier**
- Lowest portfolio variance: $\sigma^2 = 0.000036$
- Strongest backtesting results across all market regimes

Traditional + Bond Portfolio

- Delivered **stable and low volatility** performance
- Strong **risk control**: low **semi-variance** and **CVaR**
- Closest in performance to a **fully diversified** portfolio
- Best option for investors aiming for **simplified diversification with stability**

Traditional + ESG Portfolio

- Shows **notable improvements** in Omega Ratio and Sharpe Ratio compared to the traditional portfolio.
- **Tail risk is better managed**, with improved return distribution behavior post optimization.
- Provides a **balanced trade-off** between performance and responsible investing goals.
- Suitable for investors who want **ethical exposure** with moderate performance enhancement.
- Performs well in **volatile environments** due to ESG resilience factors.

8.2. Indian Market Portfolio Comparison Summary

Table 35. Indian Market Portfolio Comparison Summary

Aspect	Traditional	Trad + Commodity	Trad + ESG	Trad + Bond	All Asset Classes
Min Variance Volatility	0.1849	0.1795	0.1790	0.1787	0.1787
Max Sharpe Return	0.0864	0.0913	0.0896	0.0904	0.0951
Max Sharpe Volatility	0.1849	0.1795	0.1790	0.1787	0.1787
Omega Ratio (Equal-weight)	0.0937	0.0993	0.0978	0.0981	0.1006
Omega Ratio (Optimized)	0.0996	0.1055	0.1041	0.1047	0.1073
Sharpe Ratio (Before opt.)	0.4636	0.4746	0.4704	0.4722	0.4751
Sharpe Ratio (After opt.)	0.8947	0.9241	0.9002	0.9157	0.9486
Omega (Before opt.)	1.2024	1.2133	1.2082	1.2104	1.2162
Omega (After opt.)	1.2362	1.2483	1.2387	1.2448	1.2591
Lagrange Duality (Variance)	0.000054	0.000034	0.000036	0.000034	0.000033
Backtesting	Moderate performance; slight improvement post-opt.	Improved drawdown and returns	Stable; no significant change	Consistent, low risk	Best overall performance
Efficient Frontier & Risk-Return Shifts	Modest recovery after dip	Upward shift during post-COVID	Little change in crisis; minor recovery	Stable; efficient shape	Steep upward shift post-COVID; best frontier
Risk Metrics	Basic volatility control	Lower CVaR and semi-var.	Slight entropy and Sharpe improvement	Best downside control	Highest risk efficiency
Return Distribution	Slightly skewed, fat tails	Symmetric, thinner tails	Improved tail behavior	Lower left-tail risk	Smooth, near-normal

Indian Market Portfolio Metrics Comparison

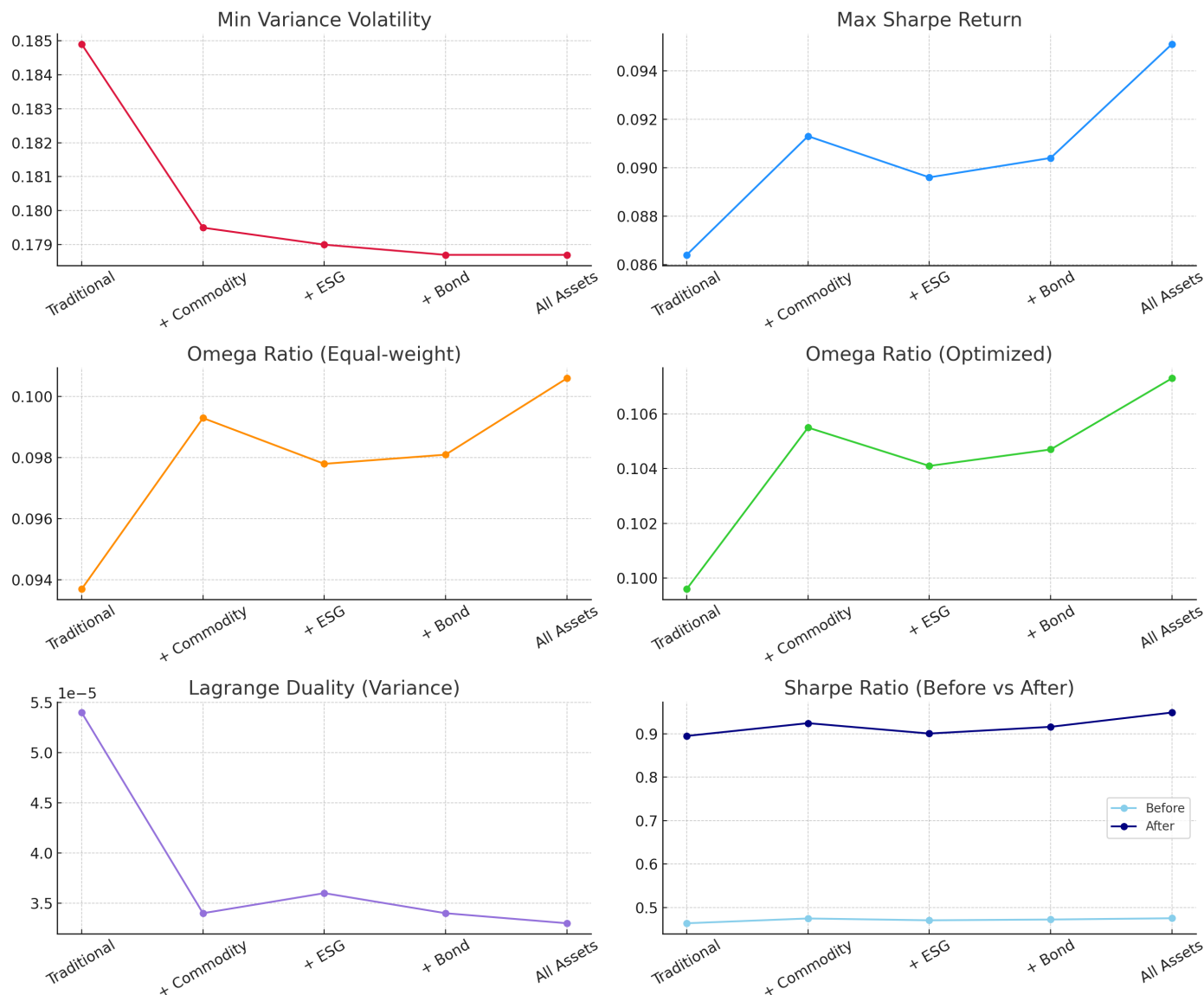


Figure 46. Line Plot Comparison of INDIAN Market Portfolio Metrics across Various Portfolio Types

8.2.1. Indian Market Portfolio Comparison Summary

- **All Asset Classes** is the best-performing portfolio across all dimensions including volatility, Sharpe ratio, Omega ratio, and Lagrange duality gap.
- **Traditional + Bond** and **Traditional + Commodity** portfolios also demonstrate meaningful improvements, making them strong intermediate options for diversification.
- Optimization significantly enhances performance in every portfolio configuration, especially when a broader set of assets is included.

Traditional + Bond Portfolio

- Offers a **very stable and risk-efficient alternative** to the fully diversified All Asset Classes portfolio.
- Performs **better than Traditional + ESG** and is **on par with Traditional + Commodity** portfolios.
- An ideal option for investors prioritizing **low volatility and moderate diversification**.
- Recommended for **risk-averse investors** who prefer **consistent returns with manageable complexity**.

Traditional + ESG Portfolio

- Demonstrates **moderate improvement** in risk-adjusted metrics, especially after optimization.
- Return distribution shows **better tail control**, though not as efficient as other combinations.
- Offers a good blend of **ethical investing** and return consistency.
- Sharpe and Omega Ratios are enhanced but remain slightly below bond and commodity combinations.
- Appeals to **sustainability-conscious investors** in emerging markets.

8.3. Europe Market Portfolio Comparison Summary

Table 36. Europe Market Portfolio Comparison Summary

Aspect	Traditional	Trad + Commodity	Trad + ESG	Trad + Bond	All Asset Classes
Min Variance Volatility	0.1906	0.1861	0.1857	0.1837	0.1830
Max Sharpe Return	0.0921	0.0973	0.0935	0.0953	0.1019
Max Sharpe Volatility	0.1906	0.1861	0.1857	0.1837	0.1830
Omega Ratio (Equal-weight)	0.0981	0.1028	0.1016	0.1032	0.1055
Omega Ratio (Optimized)	0.1035	0.1092	0.1079	0.1081	0.1108
Sharpe Ratio (Before optimization)	0.4769	0.4847	0.4823	0.4914	0.4968
Sharpe Ratio (After optimization)	0.9015	0.9462	0.9102	0.9117	0.9923
Omega (Before optimization)	1.2038	1.2156	1.2132	1.2181	1.2371
Omega (After optimization)	1.2449	1.2641	1.2463	1.2519	1.2675
Backtesting	Mild post-COVID dip; moderate recovery	Better performance during recovery	Consistent during crisis; modest recovery	Most stable return path	Strongest performance across regimes
Efficient Frontier & Risk-Return Shifts	Flattened during COVID; partial recovery	Positive shift post-COVID	Balanced shape with stable slope	Improved post-COVID slope	Strongest frontier slope post-COVID; most consistent
Risk Metrics	Basic control via volatility	Improved CVaR	Better entropy and Sharpe	Reduced semi-risk	Best CVaR, entropy, and Sharpe
Lagrange Duality (Variance)	0.000057	0.000036	0.000039	0.000037	0.000034
Return Distribution	Slightly skewed, higher tails	More centered, thinner tails	Balanced distribution	Lower left-tail risk	Well-balanced, symmetric, lowest tail risk

Europe Market Portfolio Metrics Comparison



Figure 47. Line Plot Comparison of Europe Market Portfolio Metrics across Various Portfolio Types

8.3.1. Europe Market Portfolio Comparison Summary

- **All Asset Classes Portfolio:** This portfolio consistently offers the most balanced, optimized, and risk-efficient strategy across all metrics. It achieves the lowest volatility, highest Sharpe Ratio and Omega Ratio, and shows the most stable return distribution.
- **Traditional + Bond and Traditional + Commodity Portfolios:** These combinations show meaningful improvements over the traditional portfolio alone. They provide excellent stability, improved tail risk measures, and are suitable intermediate options for investors looking for moderate diversification.
- **Portfolio Optimization:** The application of optimization techniques, including Markowitz mean-variance and risk-based metrics, consistently enhances performance. These gains are more pronounced when the portfolio includes a broader set of asset classes.
- **Recommendation:** For risk-averse investors seeking stability with manageable complexity, the **Traditional + Bond** portfolio offers a strong alternative. However, for maximum diversification and performance, the **All Asset Classes** portfolio remains the superior choice.

Traditional + ESG Portfolio

- Provides **improved entropy and Sharpe Ratio**, showing effective diversification.
- Return distribution is **more balanced** with lower tail risk than the traditional baseline.

- Post-optimization performance is **close to Traditional + Bond** configuration.
- Solid choice for investors targeting **ESG alignment and portfolio stability**.
- Performs consistently during both crisis and recovery periods, reflecting **resilience of ESG assets**.

8.4. Cross-Market Portfolio Comparison

Table 37. Market-wise Portfolio Strength Comparison

Market	Best Overall Performer	Most Stable Portfolio
U.S.	All Asset Classes (Sharpe, Omega)	Traditional + Bond (Volatility)
India	All Asset Classes (Omega, Duality)	Traditional + Bond (Sharpe, Volatility)
Europe	All Asset Classes (Sharpe, Omega)	Traditional + Bond (Volatility)

8.4.1. Cross-Market Portfolio Conclusion

- **All Asset Classes portfolio** consistently delivers the **highest total returns and risk-adjusted performance metrics** (e.g., Sharpe Ratio, Omega Ratio, and Lagrange Duality Gap) across all three markets—U.S., India, and Europe.
- **Traditional + Bond portfolio** offers a **simpler structure with lower volatility**, while still achieving well-optimized results. It strikes an effective balance between stability and return, especially suitable for investors seeking reliable diversification without complex asset exposure.
- **Portfolio selection should align with investor preferences:**
 - **Risk-seeking investors** may prefer the *All Asset Classes portfolio*, which maximizes return and diversification benefits.
 - **Risk-averse investors** or those prioritizing *consistency and lower drawdowns* may opt for the *Traditional + Bond portfolio*.

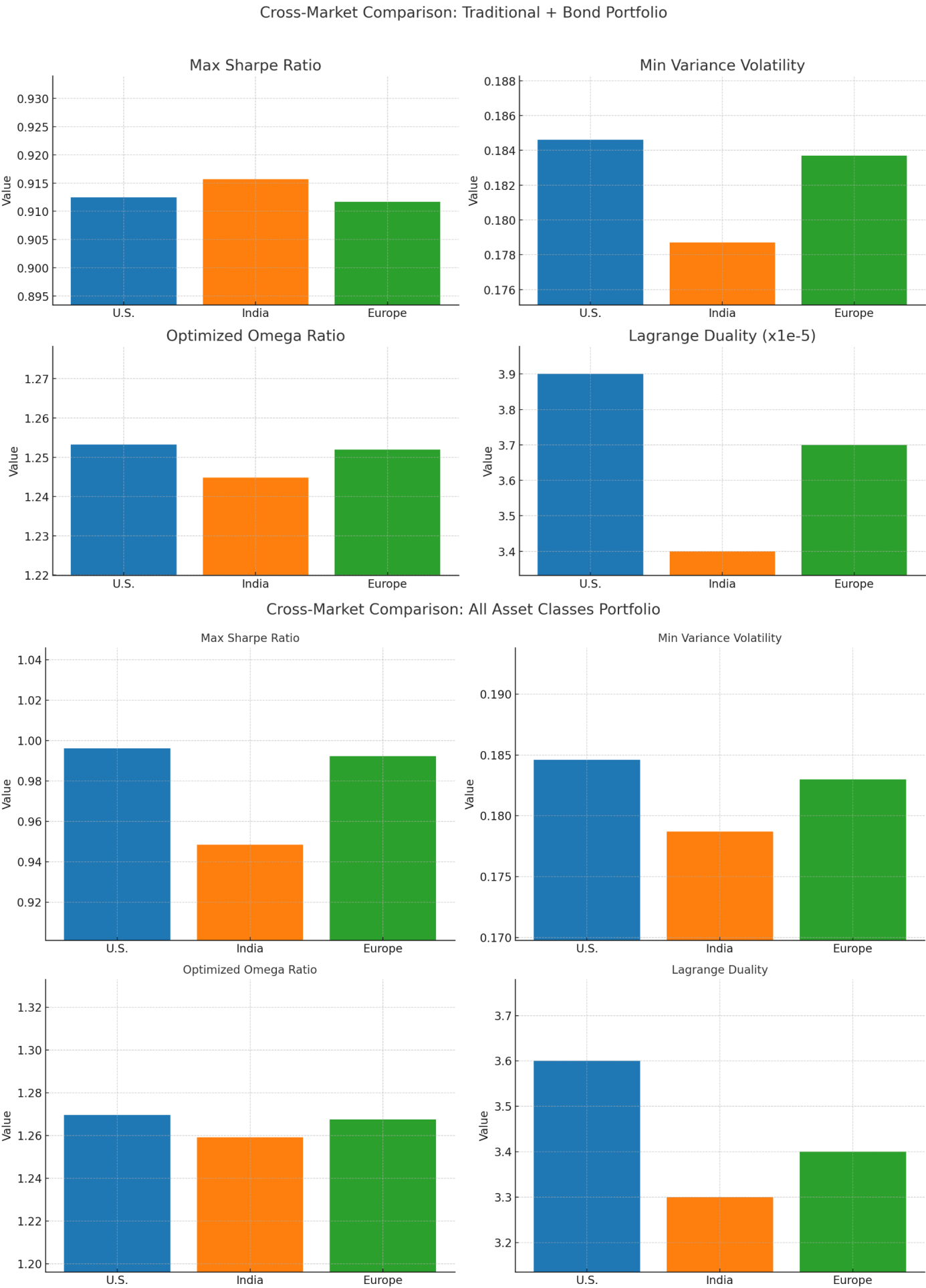


Figure 48. Comparison of Markets

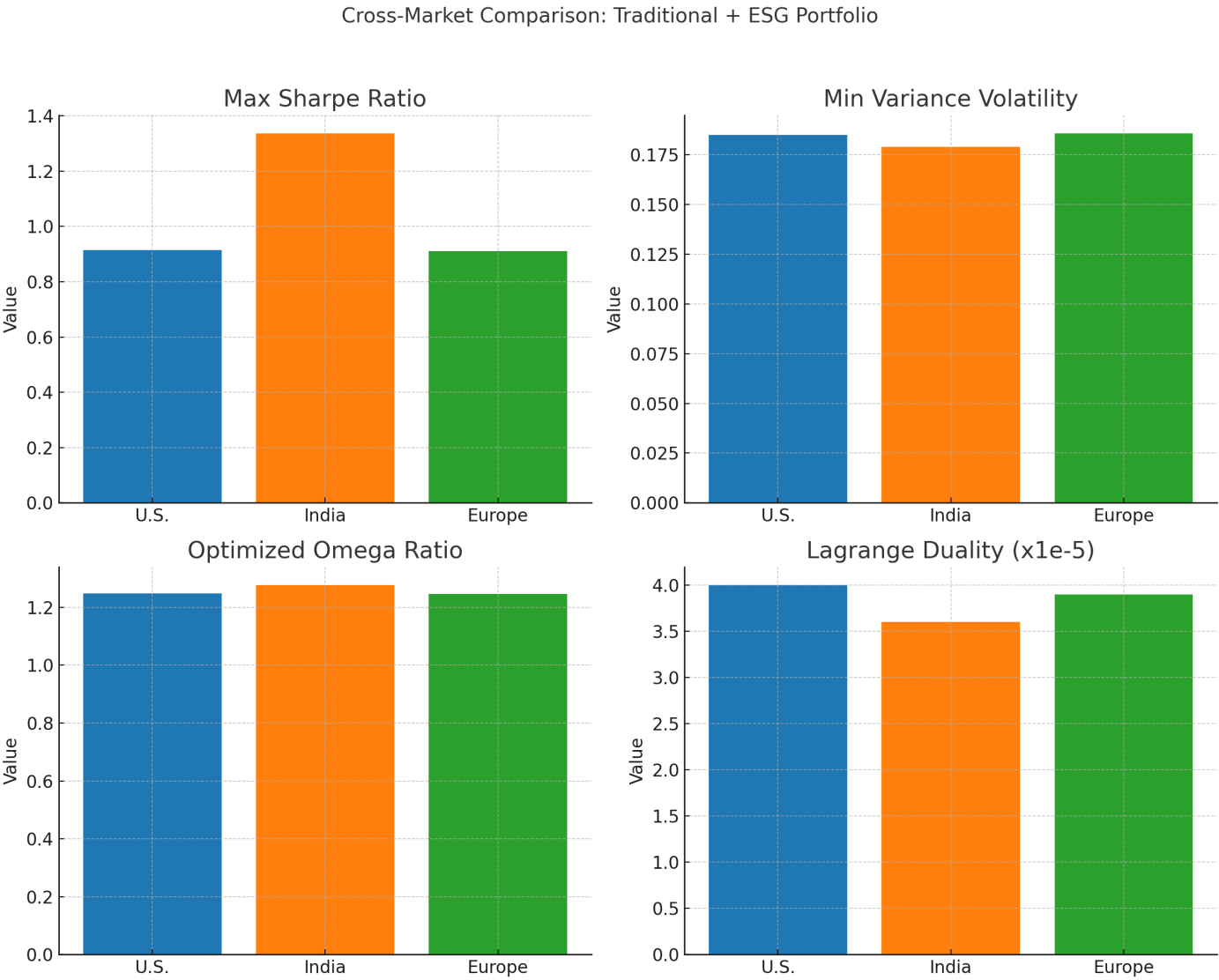


Figure 49. Comparison of Markets